Dynamic Oligopoly and Innovation

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Abstract

We develop and estimate a many-firm endogenous growth model in which firms make forward-looking R&D decisions while interacting through empirically measured product-market rivalry and technology-spillover networks. The model's linear-quadratic structure reduces the dynamic oligopoly to a coupled system of Riccati equations, which we solve for the 2017 cross section of 757 publicly listed U.S. patenting firms. Gaps between social and private returns to R&D are large and heterogeneous: the median social-to-private return ratio is 1.31. A planner who controls R&D while firms remain Cournot competitors raises aggregate R&D to 231% of the competitive level, yielding a consumption-equivalent welfare gain of 0.50%. Despite wedge heterogeneity, a 33% uniform R&D subsidy delivers a 0.45% welfare gain, because the constrained planner's gain comes mainly from expanding aggregate R&D rather than reallocating it. By contrast, a full planner that jointly controls product-market quantities and R&D raises welfare by 8.97%, highlighting the quantitative importance of product-market allocation and its interaction with innovation incentives.

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1 Introduction

Product-market competition and research and development (R&D) allocations are jointly determined. Firms' R&D decisions shape future productivity, product quality, and competitive positions, while current product-market rivalry feeds back into profits and the incentives to invest in R&D. The network-mediated part of these interactions operates through two pairwise channels: business stealing, which depresses competitors' profits, and technology spillovers, which raise other firms' productivity. In addition, firms do not generally appropriate the full household value generated by their innovations. Because these forces are heterogeneous across firms and firm pairs, the social-private R&D wedge is firm specific.

The two networks need not coincide: firms that compete closely in product markets may be distant in technology space, and firms that generate large spillovers may not be close product-market rivals. Yet common innovation-policy instruments, such as broad R&D subsidies and tax credits, change the price of R&D by the same proportion for every firm. The central question is which innovation margins such instruments can address, and which margins require information about firms' positions in the product-market rivalry and technology-spillover networks.

Existing quantitative endogenous-growth frameworks do not yet combine forward-looking strategic interaction among many firms with empirically measured product-market rivalry and technology-spillover networks at this scale. The obstacle is computational: dynamic R&D makes the state space grow rapidly with the number of interacting firms. This combination has so far limited quantitative measurement of firm-level social-private R&D wedges in large firm networks.

This paper develops and estimates an endogenous growth model in which hundreds of firms make forward-looking R&D decisions while interacting through two empirically measured networks: a product-market rivalry network and a technology-spillover network. The model makes this environment tractable by exploiting a linear-quadratic structure. Equilibrium is characterized by a coupled system of algebraic Riccati equations rather than a high-dimensional dynamic program, allowing us to solve the game for the 2017 cross section of 757 U.S. publicly listed patenting firms while retaining a continuous state space and the full firm-to-firm network matrices. The estimated model delivers firm-level social-private R&D wedges and a welfare decomposition. Despite large and heterogeneous wedges, a uniform R&D subsidy captures most of the gain available to a planner who controls R&D alone. The substantially larger gain under the full social-planner allocation comes from jointly reallocating product-market quantities and R&D effort.

The model extends Pellegrino (2025)'s static product-market framework into a dynamic setting with knowledge accumulation, R&D effort, and technology spillovers. Conditional on the current knowledge-capital vector, a static product-market game under generalized hedonic linear demand maps firms' knowledge capital into quantities and profits; its key property is that gross profits are quadratic in knowledge capital. Knowledge capital accumulates linearly through own R&D effort and through the technology-spillover network. The dynamic R&D game is therefore linear-quadratic: value functions are quadratic in the state, Markov strategies are linear in it, and computation scales polynomially rather than exponentially in the number of firms. In the deterministic version, a technology transition matrix summarizes spillovers, obsolescence, and endogenous R&D and governs the evolution of the knowledge distribution.

We estimate the framework on U.S. publicly listed firms with patents over 1989–2019. Following Pellegrino (2025), we measure product-market rivalry using the product-similarity data of Hoberg and Phillips (2016); following Bloom et al. (2013) and Lucking et al. (2019), we measure technological proximity from patent-classification overlap using the DISCERN patent-firm link, which maps U.S. publicly listed firms to their patents (Arora et al., 2024). Given these networks, the model's static equilibrium conditions recover firm-level knowledge capital from Compustat accounting data. A conditional panel relationship in recovered knowledge capital disciplines the spillover-intensity scale. The R&D efficiency and depreciation parameters are jointly calibrated to aggregate R&D intensity and instantaneous 2017 output growth. The model also reproduces untargeted cross-sectional moments: the correlation between model-implied and observed log sales is 0.96, and the corresponding correlation for log R&D expenditure is 0.72.

The estimated wedges are large and heterogeneous. At the observed 2017 knowledge-capital distribution, the median ratio of social to private marginal returns to R&D is 1.31, and social returns exceed private returns for 82.3% of firms with positive private and social marginal returns, consistent with aggregate underprovision. The dispersion is substantial: 22.5% of firms have ratios above 1.5, while for firms in the bottom tail the local first-order-condition measure points toward taxing rather than subsidizing marginal R&D. How much of the attainable R&D-policy gain a uniform instrument can capture is therefore a quantitative question. An exact flow-payoff decomposition separates each firm's wedge into four discounted components: consumer surplus, labor income, a rival-profit component operating through business stealing, and a closed-loop rival R&D resource-cost component. Consumer surplus and labor income together constitute the appropriability gap—household value generated by innovation but not captured in aggregate producer gross profit. Quantitatively, cross-firm variation in the wedge is driven mainly by consumer surplus and labor income. The negative rival-profit component substantially offsets their levels but varies much less across firms, while the rival R&D resource-cost component is small.

To separate the policy margins, we compare the competitive equilibrium with planner and monopoly counterfactuals. A constrained planner retains Cournot product-market conduct—the competitive mapping from knowledge capital to quantities—and chooses only the dynamic R&D effort rule. At the observed 2017 state, this raises aggregate R&D resource cost to about 231% of the competitive level, increases instantaneous output growth by 0.28 percentage points, and raises consumption-equivalent welfare by 0.50%. The corresponding constrained monopolist internalizes the effects of R&D on rival producers but not the consumer surplus and labor income generated by innovation. It reduces aggregate R&D resource cost to roughly one half of the competitive level and lowers welfare by 1.09%. This constrained case can also be read as R&D-only common control, or an economy-wide research joint venture. Allowing product-market quantities and R&D effort to be reallocated jointly yields a much larger welfare gain under the full planner.

We then evaluate an implementable policy: a uniform R&D subsidy financed by lump-sum taxes. Household welfare peaks at a subsidy rate of 33%, which raises instantaneous output growth by 0.27 percentage points, compared with 0.28 under the constrained planner, and welfare by 0.45%, compared with 0.50%. Despite wedge heterogeneity, a uniform instrument captures roughly ninety percent of the gain attainable through R&D policy alone because expanding aggregate R&D, rather than reallocating effort across firms, accounts for most of the constrained planner's gain.

This paper contributes to three strands of the literature. First, it contributes to work on innovation incentives, appropriability, business stealing, technology spillovers, and innovation networks (Arrow, 1962; Spence, 1984; d'Aspremont and Jacquemin, 1988; Aghion et al., 2001, 2005; Acemoglu and Akcigit, 2012; Peters, 2020; Akcigit and Ates, 2021, 2023; De Ridder, 2024). Bloom et al. (2013) provide the closest empirical evidence for this mechanism: they show that product-market proximity and technological proximity have opposite effects on innovation incentives. On innovation policy in networks, König et al. (2019) analyze R&D alliances that lower production costs while firms compete in product markets and characterize welfare-maximizing R&D subsidies. Liu and Ma (2026) study how innovation networks shape the allocation of R&D across connected firms, and Cavenaile et al. (2026) develop an oligopolistic general-equilibrium Schumpeterian model with strategic interaction among a small number of firms. Our model combines product-market rivalry and technology spillovers in a many-firm dynamic oligopoly estimated with firm-level and firm-pair-level data. It allows us to quantify the distribution of social-private

R&D wedges, connect them to welfare, and distinguish the gains attainable with uniform subsidies from those that require product-market reallocation.

Second, the paper relates to the dynamic oligopoly literature in empirical industrial organization (Ericson and Pakes, 1995; Besanko and Doraszelski, 2004; Doraszelski and Pakes, 2007; Weintraub et al., 2008). This literature develops Markov perfect industry dynamics and methods for handling large state spaces. We differ by exploiting a linear-quadratic structure that makes a full-scale many-firm continuous-state R&D game tractable with empirically measured product-market rivalry and technology-spillover networks.

Third, the paper contributes to research that incorporates oligopolistic competition into macroeconomic models (Neary, 2003; Atkeson and Burstein, 2008; Azar and Vives, 2021; Pellegrino, 2025; Ederer and Pellegrino, 2026; Bustamante and Pellegrino, 2026). Our product-market environment builds on Pellegrino (2025) and Ederer and Pellegrino (2026). Closely related contemporaneous work by Bustamante and Pellegrino (2026) develops a Network-Q model of corporate investment in oligopolistic product markets using the same Generalized Hedonic Linear (GHL) demand system. Their focus is neoclassical capital investment, firm valuation, markups, and merger counterfactuals; we instead add a technology-spillover network and forward-looking R&D investment to study how product-market rivalry and technology spillovers jointly shape growth, welfare, and innovation policy.

The remainder of the paper proceeds as follows. Section 2 constructs an endogenous growth model incorporating product-market rivalry and technology-spillover networks. Section 3 describes the data and estimation strategy. Section 4 reports the wedge measures, counterfactual allocations, and the uniform-subsidy evaluation. Section 5 concludes.

2 Model

2.1 Environment

Our environment incorporates two networks: a product-market rivalry network and a technology-spillover network. There are n oligopolistic firms, indexed by $i \in \{1, 2, \ldots, n\}$; they choose output and dynamic R&D effort while interacting strategically through these networks. We interpret each firm as supplying one composite differentiated product, so firm and product indices coincide throughout. The set of firms, product-characteristic vectors, and technological-proximity matrix are fixed over time. Innovation changes firms' knowledge capital, labor productivity, and product quality, but not their locations in product or technology space. Time is continuous and infinite, $t \in [0, \infty)$.

2.1.1 Demand

We use the GHL demand system of Pellegrino (2025) to model product-market rivalry among many oligopolistic firms. In reduced form, the final-good aggregator is

$$Y_t = \boldsymbol{q}_t^T \boldsymbol{b}_t - \frac{1}{2} \boldsymbol{q}_t^T \boldsymbol{\Sigma} \boldsymbol{q}_t \tag{1}$$

where $\boldsymbol{q}_t \in \mathbb{R}^n$ is the vector of product quantities and $\boldsymbol{b}_t \in \mathbb{R}^n$ is the vector of product-level qualities. The product-rivalry matrix is

$$\boldsymbol{\Sigma} \equiv \alpha \boldsymbol{S} + (1 - \alpha) \boldsymbol{I},$$

where $\boldsymbol{S}$ is product similarity, $\boldsymbol{I}$ is the identity matrix, and $\alpha \in [0, 1]$ maps similarity into substitutability. We normalize $S_{ii} = 1$, so $\sigma_{ii} = 1$ and $\sigma_{ij} = \alpha S_{ij}$ for $i \neq j$; thus σ_{ij} indexes pairwise product-market rivalry. Section A.1 gives the characteristic-level construction.

2.1.2 Technology

Each firm has a linear production technology in labor:

$$q_{i,t} = a_{i,t} l_{i,t} \tag{2}$$

where $a_{i,t}$ is the labor productivity of firm i at time t and $l_{i,t}$ is the production labor employed by firm i at time t.

Firms possess their own knowledge capital and use it to improve labor productivity and product quality. As shown in Equation (1), $b_{i,t}$ is the value (in final-good units) generated by the first unit of product i, so we interpret $b_{i,t}$ as product quality. Firms allocate knowledge capital between labor productivity $a_{i,t}$ and product quality $b_{i,t}$:

$$\zeta a_{i,t} + b_{i,t} = z_{i,t} \tag{3}$$

where $\zeta > 0$ governs the trade-off between improving labor productivity and product quality[1].

We model the law of motion for knowledge capital, which rises through technology spillovers from technologically similar firms and through the firm's own R&D effort. Spillovers operate through source firms' knowledge capital rather than directly through

[1]This allocation supports balanced growth; related two-margin formulations appear in Grossman et al. (2017) and Jones and Liu (2024).

their contemporaneous R&D expenditures. Let $\tilde{\mathbf{\Omega}} \in \mathbb{R}^{n\times n}$ denote recipient-normalized technology-spillover exposure, with entries $\tilde{\omega}_{ij}$. We use a row-receiver convention: row i describes the sources of spillovers received by firm i. Thus, $\tilde{\omega}_{ij} \geq 0$ is the weight placed on source firm j in firm i's spillover exposure. We set the diagonal to zero, $\tilde{\omega}_{ii} = 0$, and rows with positive off-diagonal exposure are normalized so that $\sum_{j\neq i} \tilde{\omega}_{ij} = 1$. The effective spillover matrix is $\mathbf{\Omega} = \beta\tilde{\mathbf{\Omega}}$, where $\beta > 0$ scales these relative exposure weights into spillover intensities.

Let $\boldsymbol{x}_t = (x_{1,t}, \ldots, x_{n,t})^T \in \mathbb{R}^n$ denote R&D efforts, and let $\boldsymbol{W}_t = (W_{1,t}, \ldots, W_{n,t})^T$ denote independent standard Wiener processes across firms. The control $x_{i,t}$ is an R&D effort that raises knowledge capital linearly, while the resource cost of generating this effort is quadratic, as specified in the resource constraint below. Knowledge capital follows a multi-dimensional geometric Brownian motion:

$$d\boldsymbol{z}_t = \left((\mathbf{\Omega} - \delta \boldsymbol{I})\, \boldsymbol{z}_t + \mu \boldsymbol{x}_t\right) dt + \gamma \operatorname{diag}\left(\boldsymbol{z}_t\right) d\boldsymbol{W}_t \tag{4}$$

where $\mu > 0$, $\delta > 0$, and $\gamma \geq 0$. The parameter μ governs the efficiency of R&D, γ the magnitude of shocks, and δ the depreciation rate of knowledge capital.

2.1.3 Resources and Household Preferences

We close the model with resource constraints and representative household preferences. The representative household supplies labor, owns firms, and consumes the final good. Production labor is supplied inelastically:

$$L = \sum_i l_{i,t}. \tag{5}$$

The final good is used for consumption and R&D investment. Following the endogenous growth literature (e.g. Acemoglu et al. 2016), we assume that the innovation elasticity for each firm is 0.5. Equivalently, the resource cost of firm-level R&D effort is quadratic:

$$C_t + \sum_i x_{i,t}^2 = Y_t$$

where C_t is consumption of the final good at time t. Finally, we assume that the representative household is risk-neutral and discounts the future at rate $\rho > 0$:

$$\mathbb{E}_0\left[\int_0^\infty \exp\left(-\rho t\right) C_t \, dt\right]$$

2.2 Competitive Equilibrium

We characterize a Markov perfect equilibrium with knowledge capital $\boldsymbol{z}$ as the state.

Definition 1 (Competitive equilibrium)**.** A competitive equilibrium consists of Markov R&D policies $\{x_i(\boldsymbol{z})\}_{i=1}^n$, firm value functions $\{V^i(\boldsymbol{z})\}_{i=1}^n$, static allocations $\boldsymbol{q}(\boldsymbol{z})$, $\boldsymbol{a}(\boldsymbol{z})$, $\boldsymbol{b}(\boldsymbol{z})$, the real wage $w(\boldsymbol{z})/P$, and relative prices $\boldsymbol{p}(\boldsymbol{z})/P$ such that: (i) the final-good producer maximizes profits; (ii) firms play a static Cournot equilibrium given $\boldsymbol{z}$; (iii) each firm's R&D policy is a Markov best response in the dynamic problem; (iv) knowledge capital follows Equation (4); and (v) the labor market clears.

This structure lets us separate the problem into static and dynamic components. Conditional on the current knowledge-capital vector $\boldsymbol{z}_t$, the production and technology-allocation choices determine current quantities, prices, and gross profits but do not directly affect future knowledge capital. R&D affects current payoffs through its flow cost and affects future payoffs only through the law of motion for $\boldsymbol{z}_t$. Thus, we first solve the static product-market game as a mapping from knowledge capital into profits, and then use this mapping as the payoff flow in the dynamic R&D game.

2.2.1 Final-Good Producer's Problem

The final good is produced competitively. In each period, the final-good producer chooses the quantity of each product that maximizes profits, given prices.

$$\max_{\boldsymbol{q}_t} P_t Y_t - \boldsymbol{q}_t^T \boldsymbol{p}_t$$

where $\boldsymbol{p}_t$ is the vector of product prices at time t and P_t is the final-good price at time t. The final-good producer's profit maximization problem yields the following linear inverse demand function:

$$\frac{\boldsymbol{p}_t}{P_t} = \boldsymbol{b}_t - \boldsymbol{\Sigma} \boldsymbol{q}_t \tag{6}$$

2.2.2 Static Firm Problem

We assume that firms maximize real profits, so allocations are independent of the choice of numeraire (Azar and Vives, 2021). We characterize the static Cournot-Nash equilibrium in which each firm takes the real wage and rivals' quantities as given when choosing output and the allocation of knowledge capital between labor productivity and product quality.

The static block maps the knowledge-capital vector into quantities:

$$\boldsymbol{q}_t = \boldsymbol{N}\boldsymbol{z}_t \tag{7}$$

where

$$\boldsymbol{N} \equiv \left(2\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{I}\right)^{-1},$$

and $\boldsymbol{J}$ is an $n \times n$ matrix whose entries are all one. The first term inside the inverse, $2\zeta\boldsymbol{J}/L$, captures the labor-cost channel, while $\boldsymbol{\Sigma}$ captures how quantities affect prices through product-market rivalry. The additional $\boldsymbol{I}$ is the standard Cournot own-price effect: because each firm recognizes that expanding its own quantity lowers its own price, the own-quantity term enters once more in the first-order condition. The derivation is in Section A.2. Lemma 1 shows that the inverse is well defined and that the static Cournot block is uniquely pinned down for each knowledge-capital vector.

Let $\pi_{i,t}$ denote firm i's real gross profit before subtracting R&D costs. In this equilibrium, the normalization $\sigma_{ii} = 1$ implies that static gross profit equals $q_{i,t}^2$. Hence the static payoff entering the dynamic R&D game is quadratic in knowledge capital:

$$\pi_{i,t} = q_{i,t}^2 = \boldsymbol{z}_t^T \boldsymbol{N}_i^T \boldsymbol{N}_i \boldsymbol{z}_t \equiv \boldsymbol{z}_t^T \boldsymbol{Q}_i \boldsymbol{z}_t$$

where $\boldsymbol{N}_i$ denotes the i-th row of $\boldsymbol{N}$, and $\boldsymbol{Q}_i \equiv \boldsymbol{N}_i^T \boldsymbol{N}_i$. Because gross profits are quadratic in knowledge capital, the dynamic R&D game is linear-quadratic.

2.2.3 Dynamic R&D Game

Given a static production strategy profile, we consider a dynamic R&D game. In the dynamic game, given other players' strategies $\left\{x_{j,t}\right\}_{j \neq i, t \geq 0}$, firm i chooses R&D effort $\{x_{i,t}\}_{t \geq 0}$ to maximize the expected discounted present value of profits:

$$\max_{\left\{x_{i,t}\right\}_{t \geq 0}} \quad V^i(\boldsymbol{z}_0) \equiv \mathbb{E}_0\left[\int_0^\infty \exp\left(-\rho t\right)\left\{\pi_{i,t} - x_{i,t}^2\right\} dt\right]$$

while the state evolves according to Equation (4). We use the fact that the interest rate is pinned down by the household's Euler equation, $r_t = \rho$. In firm i's Hamilton–Jacobi–Bellman (HJB) equation, x_i is the choice variable and the other components of $\boldsymbol{x}$ are evaluated at rivals' Markov strategies. The dynamic problem gives the following HJB equation:

$$\rho V^i(\boldsymbol{z}) = \max_{x_i} \left\{ \boldsymbol{z}^T \boldsymbol{Q}_i \boldsymbol{z} - x_i^2 + V_z^i(\boldsymbol{z}) \left[(\boldsymbol{\Omega} - \delta \boldsymbol{I}) \boldsymbol{z} + \mu \boldsymbol{x} \right] \right.$$
$$\left. + \frac{\gamma^2}{2} \mathrm{Tr} \left[\mathrm{diag}(\boldsymbol{z}) V_{zz}^i(\boldsymbol{z}) \mathrm{diag}(\boldsymbol{z}) \right] \right\}. \tag{8}$$

The first two terms are firm i's net instantaneous profit, while the last two terms capture expected changes in continuation value from knowledge accumulation and idiosyncratic shocks.

We seek a stabilizing equilibrium with quadratic value functions,

$$V^i(\boldsymbol{z}) = \boldsymbol{z}^T \boldsymbol{X}^i \boldsymbol{z}. \tag{9}$$

where $\boldsymbol{X}^i$ is symmetric. Let $\boldsymbol{X}_i^i$ denote its i-th column. The first-order condition gives the linear Markov strategy

$$x_i = \left(\mu \boldsymbol{X}_i^i \right)^T \boldsymbol{z}. \tag{10}$$

The derivative algebra is in Section A.4.

Stacking the feedback coefficients gives

$$\widetilde{\boldsymbol{X}} \equiv \begin{bmatrix} \boldsymbol{X}_1^1 & \cdots & \boldsymbol{X}_n^n \end{bmatrix}^T$$

and the closed-loop law of motion is

$$d\boldsymbol{z}_t = \boldsymbol{\Phi} \boldsymbol{z}_t dt + \gamma\, \mathrm{diag}(\boldsymbol{z}_t)\, d\boldsymbol{W}_t$$

where the drift matrix $\boldsymbol{\Phi}$ is defined by

$$\boldsymbol{\Phi} \equiv \boldsymbol{\Omega} - \delta \boldsymbol{I} + \mu^2 \widetilde{\boldsymbol{X}}.$$

We refer to $\boldsymbol{\Phi}$ as the *technology transition matrix*; it combines spillovers, depreciation, and endogenous R&D.

For any square matrix $\boldsymbol{A}$, let $\mathcal{D}(\boldsymbol{A})$ denote the diagonal matrix formed from its diagonal entries. With independent multiplicative shocks, the diffusion term contributes $\gamma^2 \mathcal{D}(\boldsymbol{X}^i)$ to the quadratic coefficient. Substitution into the HJB equation yields the coupled algebraic

Riccati equations

$$0 = \boldsymbol{Q}_i - \mu^2 \boldsymbol{X}_i^i \left(\boldsymbol{X}_i^i\right)^T + \left(\boldsymbol{\Phi} - \frac{\rho}{2}\boldsymbol{I}\right)^T \boldsymbol{X}^i + \boldsymbol{X}^i \left(\boldsymbol{\Phi} - \frac{\rho}{2}\boldsymbol{I}\right) + \gamma^2 \mathcal{D}\left(\boldsymbol{X}^i\right) \tag{11}$$

for $i = 1, 2, \ldots, n$. The system is coupled through $\boldsymbol{\Phi}$ and firms' feedback coefficients.

Definition 2. The solution of the stacked algebraic Riccati Equation (11) is *stabilizing* if it delivers finite expected discounted payoffs and satisfies the transversality condition

$$\lim_{T\to\infty} \mathbb{E}_0 \left[\exp(-\rho T) V^i(\boldsymbol{z}_T)\right] = 0$$

for all firms i and admissible initial states.

We retain the shock term in the model equations to state the stochastic linear-quadratic problem, although all quantitative exercises set $\gamma = 0$. In this deterministic case, stability is equivalent to all eigenvalues of $\boldsymbol{\Phi} - \rho\boldsymbol{I}/2$ having negative real parts. With multiplicative shocks, the same definition requires the discounted second moments induced by the closed-loop process to be finite. This is the stochastic counterpart of the standard growth-theory condition that discounting be sufficiently strong for infinite-horizon utility to be finite.

Assumption 1. *There exists a stabilizing solution* $\left(\boldsymbol{X}^1, \boldsymbol{X}^2, \ldots, \boldsymbol{X}^n\right)$ *of the stacked algebraic Riccati Equation* (11)*.*

Assumption 1 imposes existence, but not uniqueness, of a stabilizing solution; Section C.4 describes the numerical selection rule used in the quantitative analysis (Engwerda, 2005).

Theorem 1 (Verification of the competitive feedback rule)**.** *Fix a solution* $\{\boldsymbol{X}^j\}_{j=1}^n$ *of Equation* (11)*. Suppose the induced closed-loop process is stabilizing in the sense of Definition 2. For each firm i, take rivals' feedback rules* $x_j(\boldsymbol{z}) = \mu(\boldsymbol{X}_j^j)^T \boldsymbol{z}$ *as given, where* $\boldsymbol{X}_j^j$ *denotes the jth column of* $\boldsymbol{X}^j$*. Then*

$$x_i(\boldsymbol{z}) = \mu(\boldsymbol{X}_i^i)^T \boldsymbol{z}$$

is firm i's optimal response among admissible controls for which the state equation is well defined, the expected discounted quadratic payoff is finite, and the boundary term satisfies

$$\liminf_{T\to\infty} \mathbb{E}_0 \left[e^{-\rho T} V^i(\boldsymbol{z}_T)\right] \geq 0.$$

Therefore, conditional on existence and stabilization, a solution of the stacked Riccati system induces a linear Markov perfect equilibrium in the stated admissible class.

The proof is in Section A.4.

2.2.4 Deterministic Balanced Growth Path

Set $\gamma = 0$ and consider the closed-loop technology system $\dot{\boldsymbol{z}}_t = \boldsymbol{\Phi} \boldsymbol{z}_t$. If $\boldsymbol{\Phi}$ has a simple real dominant eigenvalue $g > 0$ with a strictly positive right eigenvector $\bar{\boldsymbol{z}}$, then $\boldsymbol{z}_t = e^{gt} c \bar{\boldsymbol{z}}$ is a balanced growth path (BGP). Thus, the long-run growth rate is the dominant eigenvalue of the technology transition matrix, and the BGP cross-section is the associated eigenvector.

Because the static mapping and R&D policies are linear in $\boldsymbol{z}_t$, quantities and R&D grow at rate g along this path. Output, consumption, and profits grow at rate $2g$ because they are quadratic in knowledge capital. The formal spectral condition, proof, growth-rate table, and partial-equilibrium diagram are collected in Section A.6. The counterfactuals below start from the observed 2017 knowledge-capital distribution, so this balanced-growth-path characterization is a long-run implication of the solved dynamic system rather than the basis for the main quantitative comparisons.

2.2.5 Household Welfare

Once the equilibrium is solved, household welfare can be computed for any initial knowledge-capital distribution with minimal additional cost. Let $V(\boldsymbol{z}_0)$ denote household welfare in the competitive equilibrium:

$$V(\boldsymbol{z}_0) = \mathbb{E}_0 \left[\int_0^\infty \exp(-\rho t) \left\{ Y_t - \sum_i x_{i,t}^2 \right\} dt \middle| \boldsymbol{z}_0 \right].$$

Total output is

$$Y_t = \boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t$$

where

$$\boldsymbol{Q} = \boldsymbol{N}^T \left(\frac{\zeta}{L} \boldsymbol{J} + \frac{1}{2} \boldsymbol{\Sigma} + \boldsymbol{I} \right) \boldsymbol{N}$$

(see Section A.3 for the derivation). The matrix $\boldsymbol{Q}$ is the output-flow matrix, not the sum of the firm payoff matrices: $\sum_i \boldsymbol{Q}_i = \boldsymbol{N}^T \boldsymbol{N}$. Their difference satisfies

$$\boldsymbol{Q} - \boldsymbol{N}^T \boldsymbol{N} = \frac{1}{2} \boldsymbol{N}^T \boldsymbol{\Sigma} \boldsymbol{N} + \frac{\zeta}{L} \boldsymbol{N}^T \boldsymbol{J} \boldsymbol{N},$$

the sum of consumer surplus and labor income. It is the household flow value not captured in aggregate producer gross profit, which we refer to as the appropriability gap. After firms' equilibrium policies are fixed, the remaining valuation equation contains no controls. Because the closed-loop process and the payoff are linear-quadratic, household welfare has the form $V(z) = z^T X z$. Substituting this quadratic form into the valuation equation gives the Lyapunov equation

$$0 = Q - \mu^2 \widetilde{X}^T \widetilde{X} + X \left(\Phi - \frac{\rho}{2} I \right) + \left(\Phi - \frac{\rho}{2} I \right)^T X + \gamma^2 \mathcal{D}(X) \tag{12}$$

Therefore, X is obtained as the solution of Equation (12), and welfare in the competitive equilibrium is $V(z_0) = z_0^T X z_0$. The valuation equation and producer-value analogue are derived in Section A.5.

2.2.6 Producer Value

We also characterize aggregate producer value in the competitive equilibrium. This object provides a measure of producers' dynamic value that can be compared with household welfare and with the counterfactual allocations below. Aggregate real profit is

$$\sum_i \pi_{i,t} = q_t^T q_t = z_t^T N^T N z_t.$$

Relative to the household-welfare calculation in Section 2.2.5, aggregate producer value is obtained by solving a modified version of Equation (12) in which Q is replaced with $N^T N$. Denoting the solution by X^P, aggregate producer value is $V^P(z_0) = z_0^T X^P z_0$.

2.3 Planner, Monopoly, and Constrained Allocations

We use counterfactual allocations to decompose distortions along two margins: the static product-market mapping and the controller of the dynamic R&D effort rule. The scenario labels are two-letter codes: the first letter denotes the static product-market mapping, and the second denotes the R&D-effort controller. The label C denotes competition, S the social planner, and M monopoly. Thus, CC is the decentralized benchmark: firms produce according to the Cournot mapping $q = Nz$, and each firm chooses its own R&D effort to maximize its own value. We use CC+s for the corresponding equilibrium under a uniform R&D subsidy at rate $s \in [0, 1)$, which reduces each firm's private R&D cost from x_i^2 to $(1 - s)x_i^2$ and is financed by lump-sum taxes. Table 1 summarizes the allocations defined in this section.

2.3.1 Social Planner

In scenario SS, the planner chooses product-market allocations and R&D effort to maximize household welfare subject to the economy's resource and technological constraints.

Table 1: Counterfactual Allocations in the Model

Scenario	Static mapping	R&D-effort controller	Flow payoff before R&D resource cost
CC	$\boldsymbol{q} = \boldsymbol{N}\boldsymbol{z}$	Firm i maximizes own value	$\boldsymbol{z}^T\boldsymbol{Q}_i\boldsymbol{z}$
CS	$\boldsymbol{q} = \boldsymbol{N}\boldsymbol{z}$	Constrained planner maximizes household welfare	$\boldsymbol{z}^T\boldsymbol{Q}\boldsymbol{z}$
CM	$\boldsymbol{q} = \boldsymbol{N}\boldsymbol{z}$	Monopolist maximizes aggregate producer value	$\boldsymbol{z}^T\boldsymbol{N}^T\boldsymbol{N}\boldsymbol{z}$
CC+s	$\boldsymbol{q} = \boldsymbol{N}\boldsymbol{z}$	Firm i maximizes own value with private R&D cost $(1-s)x_i^2$	$\boldsymbol{z}^T\boldsymbol{Q}_i\boldsymbol{z}$
SS	$\boldsymbol{q} = \boldsymbol{N}^*\boldsymbol{z}$	Social planner maximizes household welfare	$\frac{1}{2}\boldsymbol{z}^T\boldsymbol{N}^*\boldsymbol{z}$
MM	$\boldsymbol{q} = \boldsymbol{N}^M\boldsymbol{z}$	Monopolist maximizes aggregate producer value	$\boldsymbol{z}^T\boldsymbol{P}^M\boldsymbol{z}$

Notes: Each dynamic objective subtracts R&D resource costs from the flow payoff shown. In CC+s, subsidy payments are lump-sum financed, so household welfare subtracts the full resource cost $\boldsymbol{x}^T\boldsymbol{x}$. CS and CM retain the Cournot mapping $\boldsymbol{q} = \boldsymbol{N}\boldsymbol{z}$ and change only the R&D controller.

Static Problem Given $\boldsymbol{z}_t$, the planner solves

$$\max_{\boldsymbol{a}_t,\boldsymbol{b}_t,\boldsymbol{l}_t,\boldsymbol{q}_t} Y_t = \boldsymbol{q}_t^T\boldsymbol{b}_t - \frac{1}{2}\boldsymbol{q}_t^T\boldsymbol{\Sigma}\boldsymbol{q}_t$$

subject to

$$\begin{aligned} q_{i,t} &= a_{i,t}l_{i,t} \quad i = \{1,2,\ldots,n\} \\ z_{i,t} &= \zeta a_{i,t} + b_{i,t} \quad i = \{1,2,\ldots,n\} \\ L &= \sum_i l_{i,t} \end{aligned} \tag{13}$$

The solution yields quantities as linear functions of knowledge capital:

$$\boldsymbol{q}_t^* = \boldsymbol{N}^*\boldsymbol{z}_t$$

and a quadratic representation of output in knowledge capital

$$Y_t^* = \frac{1}{2}\boldsymbol{z}_t^T\boldsymbol{N}^*\boldsymbol{z}_t$$

where

$$\boldsymbol{N}^* = \left(2\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma}\right)^{-1}$$

(see Section B.1).

Dynamic Problem Given the static output coefficient, the planner chooses R&D effort to maximize

$$V^{SS}(\boldsymbol{z}_0) \equiv \max_{\{x_t\}_{t\geq 0}} \mathbb{E}_0\left[\int_0^\infty \exp(-\rho t)\left\{\frac{1}{2}\boldsymbol{z}_t^T\boldsymbol{N}^*\boldsymbol{z}_t - \boldsymbol{x}_t^T\boldsymbol{x}_t\right\}dt\right],$$

subject to the knowledge accumulation law in Equation (4). This is again a linear-quadratic control problem, so the optimal R&D effort rule is linear in knowledge capital:

$$\boldsymbol{x} = \mu\boldsymbol{X}^{SS}\boldsymbol{z} \tag{14}$$

and the induced law of motion can be written as

$$d\boldsymbol{z}_t = \boldsymbol{\Phi}^{SS}\boldsymbol{z}_t dt + \gamma\,\mathrm{diag}(\boldsymbol{z}_t)\,d\boldsymbol{W}_t$$

where

$$\boldsymbol{\Phi}^{SS} \equiv \boldsymbol{\Omega} - \delta\boldsymbol{I} + \mu^2\boldsymbol{X}^{SS}$$

Substituting the quadratic planner value and the linear R&D rule into the planner's HJB yields a single algebraic Riccati equation:

$$0 = \frac{1}{2}\boldsymbol{N}^* - \mu^2\left(\boldsymbol{X}^{SS}\right)^2 + \boldsymbol{X}^{SS}\left(\boldsymbol{\Phi}^{SS} - \frac{\rho}{2}\boldsymbol{I}\right) + \left(\boldsymbol{\Phi}^{SS} - \frac{\rho}{2}\boldsymbol{I}\right)^T\boldsymbol{X}^{SS} + \gamma^2\mathcal{D}\left(\boldsymbol{X}^{SS}\right) \tag{15}$$

Therefore, $\boldsymbol{X}^{SS}$ is obtained as the solution of Equation (15), and household welfare under the optimal allocation is $V^{SS}(\boldsymbol{z}_0) = \boldsymbol{z}_0^T\boldsymbol{X}^{SS}\boldsymbol{z}_0$. The HJB and derivative steps are in Section B.2. Unlike the competitive equilibrium, which requires one Riccati equation per firm, the social-planner problem requires only a single Riccati equation.

2.3.2 Monopoly

In scenario MM, a monopolist jointly chooses production and R&D effort to maximize aggregate producer value. In the static product-market block, it maximizes

$$\Pi_t \equiv \frac{\boldsymbol{q}_t^T\boldsymbol{p}_t}{P_t} - \frac{w_t}{P_t}\sum_i l_{i,t}.$$

Solving the monopolist's static problem yields

$$\boldsymbol{q}_t^M = \boldsymbol{N}^M \boldsymbol{z}_t, \qquad \boldsymbol{N}^M \equiv \frac{1}{2}\left(\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma}\right)^{-1},$$

and aggregate firm profit can be written as

$$\Pi_t^M = \boldsymbol{z}_t^T \boldsymbol{P}^M \boldsymbol{z}_t, \qquad \boldsymbol{P}^M \equiv (\boldsymbol{N}^M)^T \boldsymbol{\Sigma} \boldsymbol{N}^M.$$

The superscript P denotes producer value, distinguishing the monopolist's objective from household welfare under the induced policy. The monopolist's dynamic producer-value problem is therefore

$$V^{P,MM}(\boldsymbol{z}_0) \equiv \max_{\{\boldsymbol{x}_t\}_{t\geq 0}} \mathbb{E}_0 \left[\int_0^\infty \exp\left(-\rho t\right) \left\{\boldsymbol{z}_t^T \boldsymbol{P}^M \boldsymbol{z}_t - \boldsymbol{x}_t^T \boldsymbol{x}_t\right\} dt\right],$$

subject to the knowledge accumulation law. This problem is solved by replacing $\boldsymbol{N}^*/2$ with $\boldsymbol{P}^M$ in Equation (15). Let $\boldsymbol{X}^{P,MM}$ denote the solution. The monopoly R&D effort rule is then $\boldsymbol{x}_t = \mu \boldsymbol{X}^{P,MM} \boldsymbol{z}_t$. Because the monopolist maximizes producer value rather than household welfare, the household welfare reported for MM is computed separately from a Lyapunov equation using the monopoly output matrix $\boldsymbol{Q}^M$ and the technology transition induced by $\boldsymbol{X}^{P,MM}$. For the derivation of $\boldsymbol{N}^M$, $\boldsymbol{P}^M$, $\boldsymbol{Q}^M$, and the welfare equation, see Section B.3.

2.3.3 Separating Product-Market and R&D Control

Scenarios CS and CM retain the Cournot mapping $\boldsymbol{q}_t = \boldsymbol{N}\boldsymbol{z}_t$ while assigning R&D control to the planner and monopolist, respectively. Starting from $\boldsymbol{z}_0$, the constrained planner in CS maximizes

$$V^{CS}(\boldsymbol{z}_0) \equiv \max_{\{\boldsymbol{x}_t\}_{t\geq 0}} \mathbb{E}_0 \left[\int_0^\infty \exp\left(-\rho t\right) \left\{\boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t - \boldsymbol{x}_t^T \boldsymbol{x}_t\right\} dt\right],$$

subject to

$$d\boldsymbol{z}_t = \left((\boldsymbol{\Omega} - \delta \boldsymbol{I})\, \boldsymbol{z}_t + \mu \boldsymbol{x}_t\right) dt + \gamma \operatorname{diag}(\boldsymbol{z}_t)\, d\boldsymbol{W}_t.$$

Unlike in Section 2.3.1, the constrained planner's gross instantaneous return is $\boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t$ rather than $\frac{1}{2}\boldsymbol{z}_t^T \boldsymbol{N}^* \boldsymbol{z}_t$. Therefore, we obtain the constrained planner's allocation by solving a modified version of Equation (15), in which $\boldsymbol{N}^*/2$ is replaced with $\boldsymbol{Q}$. Denoting the solution of the modified Riccati equation by $\boldsymbol{X}^{CS}$, household welfare under the constrained-planner allocation is given by $V^{CS}(z_0) = z_0^T X^{CS} z_0$.

The constrained monopolist in scenario CM is defined analogously, except that the dynamic controller maximizes aggregate producer value rather than household welfare. Given the competitive static mapping, $q_t = N z_t$, aggregate real profit is $z_t^T N^T N z_t$. Hence the constrained monopolist chooses R&D effort to maximize

$$V^{P,CM}(z_0) \equiv \max_{\{x_t\}_{t\geq 0}} \mathbb{E}_0 \left[\int_0^\infty \exp(-\rho t) \left\{ z_t^T N^T N z_t - x_t^T x_t \right\} dt \right],$$

subject to the same knowledge accumulation law. This allocation is obtained by replacing $N^*/2$ with $N^T N$ in Equation (15). Let $X^{P,CM}$ denote the resulting producer-value coefficient, so the induced R&D effort rule is $x_t = \mu X^{P,CM} z_t$. Household welfare under this constrained-monopolist effort rule is then computed from the same Lyapunov equation used in Section 2.2.5, with the competitive output matrix Q and the technology transition induced by $X^{P,CM}$. Denoting the resulting welfare coefficient by X^{CM}, household welfare is $V^{CM}(z_0) = z_0^T X^{CM} z_0$. In the numerical analysis, we also compare these planner and monopoly allocations with the uniform-subsidy equilibrium CC+s.

3 Data and Estimation

We map the model's product-market rivalry and technology-spillover networks, knowledge capital, and dynamic parameters to observed data. We use a 1989–2019 panel to construct annual networks, recover firm-level knowledge capital, and discipline the spillover-intensity scale. The quantitative baseline and counterfactuals are solved at the observed 2017 cross section of 757 firms.

3.1 Sample Construction and Empirical Mapping

For each year, we retain firms covered by Compustat and the Hoberg–Phillips product-similarity data that have positive gross profits and at least ten DISCERN-linked patent-classification records in the five-year window used to measure technology shares. The resulting annual samples range from 418 firms in 1989 to 802 firms in 1998, and the 2017 quantitative baseline contains 757 firms. The 2017 baseline captures most activity in the relevant patenting-public-firm universe: among 2017 Compustat firms with positive gross profits and at least one DISCERN-linked patent in the 2015–2019 grant window, the baseline firms account for about 91% of sales, 98% of observed R&D, 94% of gross profits, and 99.5% of linked patents. Table 4 summarizes the estimated parameters and their sources.

Table 2: Mapping Data to Model Objects

Model object	Data source or empirical object	Mapping to model
$\boldsymbol{S}, \boldsymbol{\Sigma}(\alpha)$	Hoberg-Phillips product similarity	Hoberg cosine similarities measure $\boldsymbol{S}$; the model sets $\boldsymbol{\Sigma} = \alpha \boldsymbol{S} + (1-\alpha)\boldsymbol{I}$, with diagonal entries normalized to one.
$\tilde{\boldsymbol{\Omega}}$	DISCERN patents and Cooperative Patent Classification (CPC) groups	Jaffe overlaps are computed from five-year patent shares; the diagonal is set to zero and rows are normalized by recipient.
$\boldsymbol{q}$	Gross profits, Compustat `revt` minus `cogs`	The static first-order condition implies $\pi_i = q_i^2$, so quantities are recovered as $q_i = \sqrt{\pi_i}$.
ζ/L	Aggregate cost of goods sold and recovered quantities	The aggregate production-cost condition identifies the ratio ζ/L.
$\boldsymbol{z}$	Recovered $\boldsymbol{q}$, $\boldsymbol{\Sigma}$, and ζ/L	The static equilibrium condition recovers $\boldsymbol{z}_t = (2(\zeta/L)\boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{I})\boldsymbol{q}_t$.
β	Panel growth of recovered knowledge capital and R&D controls	The conditional panel relationship disciplines the spillover-intensity scale.
μ and δ	R&D expenditure and instantaneous 2017 output growth	Own-R&D productivity and depreciation are jointly calibrated in the 2017 baseline; R&D effort is endogenous in the model.

Notes: Nominal accounting variables are deflated using the GDP deflator. The effective spillover matrix is $\boldsymbol{\Omega} = \beta\tilde{\boldsymbol{\Omega}}$. Observed R&D expenditures enter the panel controls and calibration moments. The recovered 2017 knowledge-capital vector is the initial state in the numerical analysis.

Table 2 summarizes the empirical procedure: selecting the sample, constructing the two networks, recovering firm-level quantities and knowledge capital from Compustat variables, and estimating or calibrating the remaining dynamic parameters.

3.2 Product-Market Proximity and Substitutability

Following Pellegrino (2025), we measure the product-similarity matrix $\boldsymbol{S}$ using the Hoberg-Phillips 10-K text similarity data (Hoberg and Phillips, 2016). These cosine-similarity scores predict competitive relationships and provide the empirical counterpart of $\boldsymbol{\Psi}^T\boldsymbol{\Psi}$ in the GHL microfoundation in Section A.1. The numerical analysis constructs the product-rivalry matrix $\boldsymbol{\Sigma}(\alpha) = \alpha\boldsymbol{S} + (1-\alpha)\boldsymbol{I}$ before solving the static and dynamic blocks, so for $i \neq j$, $\sigma_{ij} = \alpha S_{ij}$, while the model normalization imposes $S_{ii} = \sigma_{ii} = 1$.

We take α, which governs the degree of horizontal differentiation across goods, from Pellegrino (2025). That paper disciplines α using the inverse cross-price elasticity reported by Nevo (2001) and shows that the implied price elasticities are aligned with estimates for automobiles, ready-to-eat cereal, and computers (Berry et al., 1995; Nevo, 2001; Goeree, 2008). Appendix Table 12 reports a local sensitivity exercise that recovers the 2017 knowledge-capital state and recalibrates the dynamic parameters at nearby values of α.

3.3 Technological Proximity

We construct Jaffe technological proximity from CPC-group shares in the DISCERN 2 patent-firm dataset (Bloom et al., 2013; Lucking et al., 2019; Arora et al., 2024). Patent shares use grants from $t-2$ through $t+2$ around each sample year. This five-year window reduces noise from sparse patent-class counts and partly accommodates the median three-year filing-to-grant lag in the linked patents.

Let T_i denote the vector of firm i's patent shares across technology classes. The raw Jaffe overlap (Jaffe, 1986) between firms i and j is

$$\hat{\omega}_{ij} = \frac{T_i T_j'}{\left(T_i T_i'\right)^{1/2} \left(T_j T_j'\right)^{1/2}}.$$

As in Liu and Ma (2026), we convert these raw overlaps into recipient-normalized spillover weights. We set the diagonal entries to zero and normalize each recipient firm's row:

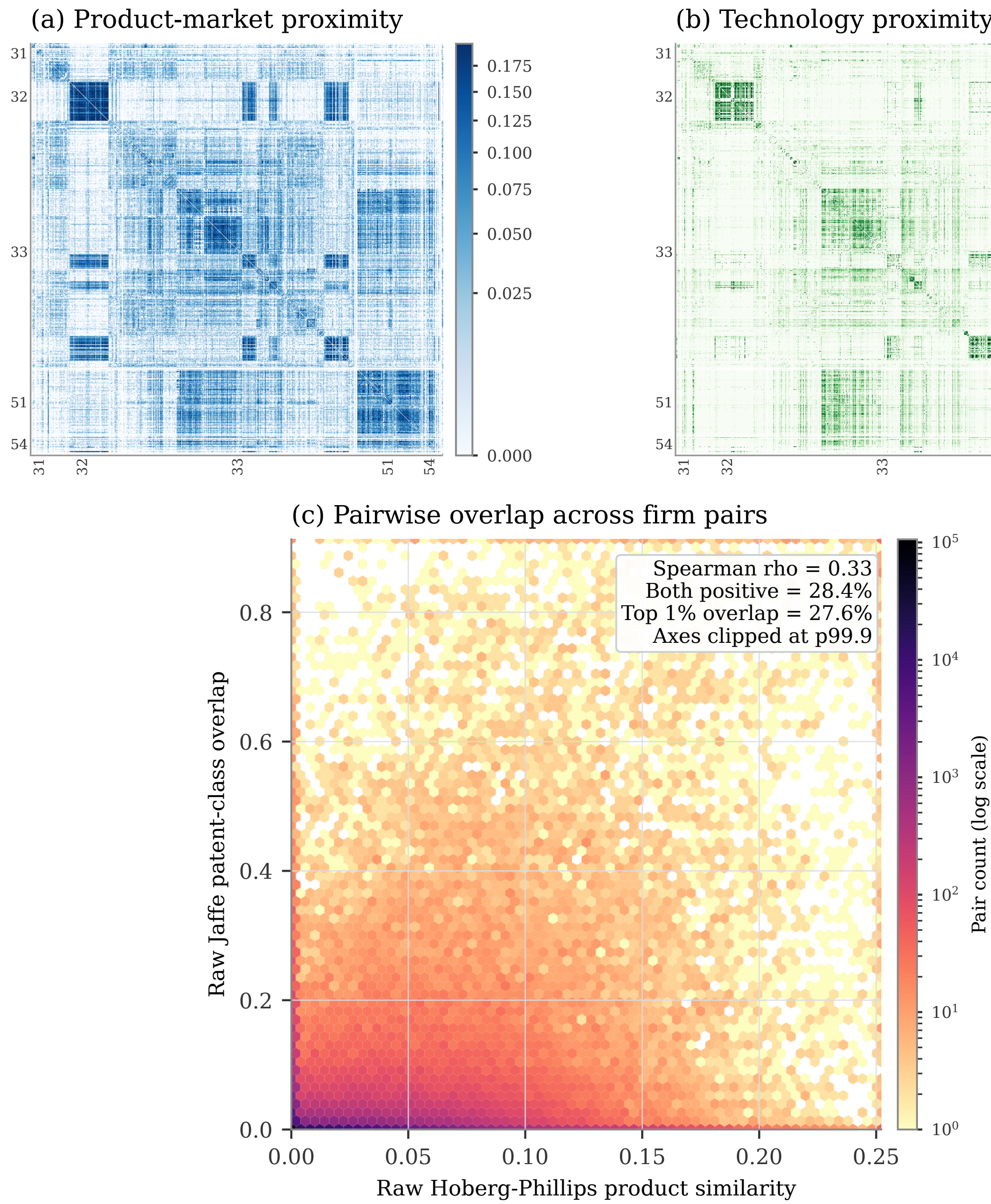
$$\tilde{\omega}_{ij} = \frac{\hat{\omega}_{ij}}{\sum_{k \neq i} \hat{\omega}_{ik}}, \qquad j \neq i.$$

Rows with no off-diagonal overlap remain zero. Thus $\tilde{\omega}_{ij}$ is source j's share of firm i's spillover exposure, and $\mathbf{\Omega} = \beta \tilde{\mathbf{\Omega}}$, with β governing total exposure. Recipient normalization generally makes $\tilde{\mathbf{\Omega}}$ asymmetric.

The product-market and technology networks are empirically distinct. To summarize their firm-level overlap, define product-market strength as the off-diagonal row sum $\sum_{j \neq i} \sigma_{ij}$ and technology-source strength as the off-diagonal column sum $\sum_{j \neq i} \Omega_{ji}$. The latter follows our row-receiver convention: it measures firm i's total weight as a spillover source across recipient firms. Across the 757 firms in the 2017 baseline, the Spearman correlation between these two strength measures is 0.22. It is 0.27 when each destination firm's contribution is weighted by its share of 2017 sample sales. Figure 1 visualizes the network distinction at the pair level. The figure uses raw product and technology proximities, before the row normalization that maps Jaffe overlap into spillover exposure.

Appendix Figure 8 shows that the product-market and technology networks are highly persistent at short horizons but decay over longer horizons. The fixed-network counterfactuals therefore describe observed-state comparisons around the 2017 network.

Figure 1: Product-Market and Technology Proximity in 2017



Notes: Panels (a) and (b) plot the 757 baseline firms, sorted by detailed North American Industry Classification System (NAICS) code and sales; boundaries mark two-digit NAICS sectors. Panel (a) reports raw Hoberg-Phillips product similarity and panel (b) raw Jaffe patent-class overlap. Panel (c) plots their joint distribution across off-diagonal firm pairs using log-count hexagons. Axes are clipped at the 99.9th percentiles for readability; reported statistics use all pairs.

3.4 Knowledge Capital Distribution

Next, we recover firm-level quantities, the ratio ζ/L, and knowledge capital $\boldsymbol{z}_t$ from Compustat accounting variables. The recovery proceeds in three steps.

First, we measure firm-level gross profit $\pi_{i,t}$ as total revenue (revt) minus cost of goods sold (cogs), with nominal variables deflated using the GDP deflator. In the model, the static Cournot first-order condition implies $\pi_{i,t} = q_{i,t}^2$. This identity is the reason we require positive gross profits in the sample: it lets us recover quantities as $q_{i,t} = \sqrt{\pi_{i,t}}$.

Second, we identify ζ/L from aggregate production costs. We interpret Compustat cost of goods sold as the empirical counterpart of variable production costs in the static block. Under this interpretation, the model implies

$$\frac{\zeta}{L}\boldsymbol{q}_t^T \boldsymbol{J} \boldsymbol{q}_t = \text{total production cost at time } t.$$

We calculate total production cost as the sum of cogs for all firms in the sample.

Third, we recover the knowledge-capital vector from the static equilibrium condition. Given the recovered quantities and product-rivalry matrix, Equation (7) implies

$$\boldsymbol{z}_t = \left(2\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{I}\right)\boldsymbol{q}_t.$$

Because $\boldsymbol{\Sigma}$ has unit diagonal, the recovery matrix contains $2\boldsymbol{I}$ on its diagonal. The recovered $\boldsymbol{z}_t$ enters the panel estimation and supplies the 2017 initial state.

3.5 Knowledge-Capital Law of Motion

We measure R&D effort as the square root of Compustat R&D expenditure, so $x_{i,t}^2$ corresponds to observed expenditure. Dividing the model's knowledge-accumulation equation by $z_{i,t}$ implies that proportional knowledge growth depends on recipient-weighted neighbor knowledge, depreciation, and own R&D intensity, $x_{i,t}/z_{i,t}$. Its annual-difference analogue motivates

$$\log z_{i,t+1} - \log z_{i,t} = \beta \sum_{j \neq i} \tilde{\omega}_{ij,t} \frac{z_{j,t}}{z_{i,t}} + \text{Controls}_{i,t} + \epsilon_{i,t} \tag{16}$$

where $z_{i,t}$ denotes firm i's knowledge capital at time t, and $\tilde{\omega}_{ij,t}$ measures the recipient-normalized technological proximity between firms i and j at time t. The controls include year and firm fixed effects, $\log z_{i,t}$, and R&D-effort intensity $x_{i,t}/z_{i,t}$; the fixed effects absorb the constant depreciation term and common growth components. The panel specification

Table 3: Knowledge-Capital Growth and Technology Exposure

	(1)	(2)	(3)
Technology exposure, $\sum_{j\neq i} \tilde{\omega}_{ij,t} z_{j,t}/z_{i,t}$	0.039***	0.021**	0.025***
	(0.010)	(0.008)	(0.008)
Product-market exposure, $\sum_{j\neq i} s_{ij,t} z_{j,t}/z_{i,t}$		-0.008***	-0.009***
		(0.002)	(0.002)
Controls: x_i/z_i, $\log z_i$	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes
Year/interacted fixed effects	Year	Year	Tech-year
Observations	18,724	18,724	18,724

Notes: The dependent variable is $\Delta \log z_i$. Technology exposure uses the recipient-normalized proximity weights $\tilde{\omega}_{ij,t}$; product-market exposure uses the Hoberg–Phillips similarity scores $s_{ij,t}$. The structural spillover matrix is $\mathbf{\Omega}_t = \beta \tilde{\mathbf{\Omega}}_t$. Standard errors, two-way clustered by firm and calendar year, are in parentheses. All columns use the full sample, ordinary least squares (OLS), exact-calendar growth, firm fixed effects, x_i/z_i, and $\log z_i$. Variables are cleaned by 0.5 percent winsorization before and after exposure construction, with $\log z_i$ trimmed. Product-market exposure is included where reported. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

conditions on own R&D intensity, while aggregate 2017 moments discipline μ and δ below. We also report specifications that add product-market-neighbor exposure to separate technology-neighbor exposure from product-market-neighbor exposure.

Table 3 reports estimates of Equation (16). Column (1) reports the raw technology-exposure association: the coefficient on $\sum_{j\neq i} \tilde{\omega}_{ij,t} z_{j,t}/z_{i,t}$ is 0.039 and statistically significant. Column (2) adds product-market-neighbor exposure, $\sum_{j\neq i} s_{ij,t} z_{j,t}/z_{i,t}$, to separate technology exposure from product-market-neighbor exposure. The technology coefficient falls to 0.021 but remains positive and statistically significant, while the product-market exposure coefficient is negative. Column (3) replaces year fixed effects with modal-technology-section-by-year fixed effects; the technology coefficient is 0.025. Standard errors are two-way clustered by firm and calendar year in all three columns.

Appendix Table 9 shows that the technology-exposure coefficient remains positive and statistically significant when technological proximity is measured using a main-group Mahalanobis measure. Appendix Table 10 shows that the same three OLS specifications remain positive when the dependent variable is cumulative three-year knowledge-capital growth, $\log z_{i,t+3} - \log z_{i,t}$; the two adjusted specifications are statistically significant at the 10% level. These cumulative coefficients provide a persistence check.

We use $\beta = 0.02$, the rounded adjusted one-year coefficient, as the central numerical anchor, with $\beta = 0.01$ and $\beta = 0.03$ as sensitivity cases.

Table 4: Parameters

Notation	Description	Value	Source
$\boldsymbol{S}$	Product similarity	–	Hoberg and Phillips (2016)
$\tilde{\boldsymbol{\Omega}}$	Technology-spillover exposure weights	–	DISCERN patents, CPC groups
α	Product similarity to rivalry	0.12	Pellegrino (2025)
β	Spillover-intensity scale	0.020	Rounded central estimate from the law of motion
γ	Std. dev. of idiosyncratic shocks	0.00	Deterministic economy in the baseline
ζ/L	Labor augmentation efficiency	0.0040	Compustat, cost of goods sold
ρ	Discount rate	0.10	Externally set benchmark
μ	R&D coefficient	0.0524	Jointly match R&D share and growth
δ	Depreciation rate	0.0112	Jointly match R&D share and growth

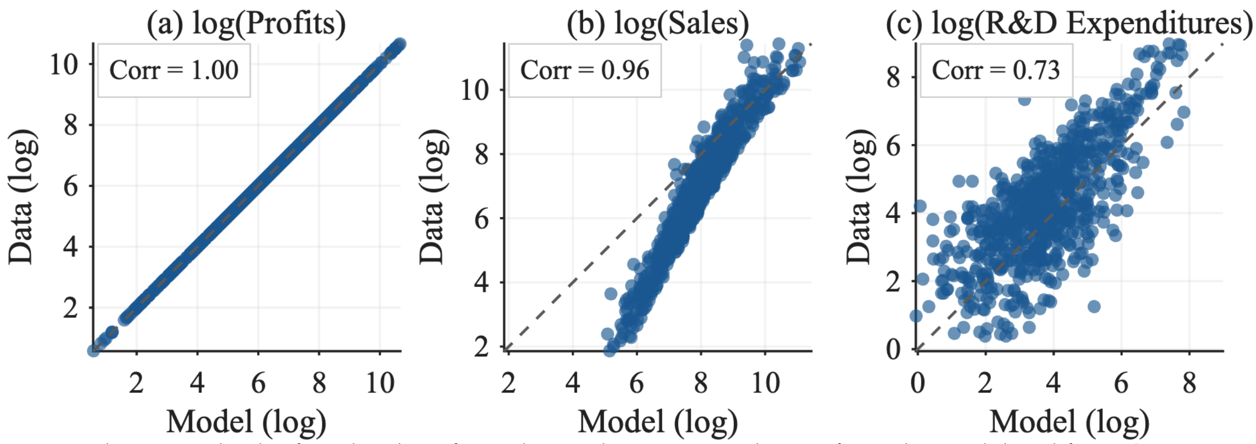
Notes: Dashes indicate matrix-valued empirical objects. The baseline sets $\gamma = 0$ for the deterministic economy. Aggregate production costs identify the ratio ζ/L. The parameters μ and δ are jointly calibrated in the 2017 competitive baseline to match aggregate R&D intensity and instantaneous 2017 g_Y.

3.6 Remaining Dynamic Parameters

With the spillover channel disciplined as above, we set the discount rate ρ externally and use aggregate moments to discipline the remaining dynamic parameters, the R&D coefficient μ and the depreciation rate δ. The baseline sets $\rho = 10\%$; Appendix Table 13 reports results for lower discount rates.

We calibrate μ and δ jointly in the 2017 competitive-equilibrium baseline by matching two aggregate moments: R&D intensity of 2.6% and average real GDP per capita growth of 1.5% over 1989–2019. We use real GDP per capita growth because Y is real final-good output and the fixed-firm model abstracts from population growth and entry; the target therefore anchors the intensive-margin growth rate rather than nominal sales growth within the selected firms. For any candidate pair (μ, δ), we solve the 2017 competitive equilibrium at the observed knowledge-capital vector and compute the model-implied R&D intensity and the instantaneous 2017 output-growth rate, $g_Y \equiv d \log Y_t / dt$. We measure model-implied R&D intensity as aggregate R&D expenditure, $\sum_i x_i^2$, divided by labor payments plus gross operating profits. The calibration chooses μ and δ to match these two targets jointly, yielding $\mu = 0.0524$ and $\delta = 0.0112$.

Figure 2: Firm-Level Profits, Sales, and R&D Expenditures: Model vs. Data



Notes: These panels plot firm-level profits, sales, and R&D expenditures from the model and from Compustat. Profits in the data are measured as "Revenue – Total" (revt) minus "Cost of Goods Sold" (cogs) in Compustat. The model is calibrated so that firm-level profits match those in the data.

3.7 Model Fit

We evaluate the model's fit using both targeted and non-targeted moments. Figure 2 compares firm-level profits, sales, and R&D expenditures in the model and in the data. Profits are targeted by construction because we recover quantities from the profit identity implied by the static equilibrium. The more informative checks are therefore sales and R&D. The model reproduces the cross-sectional distribution of sales closely, with a correlation of 0.96 between model-implied and observed log sales. It also captures a substantial part of the dispersion in innovation activity: the correlation between model-implied and observed log R&D expenditures is 0.72.

Appendix Section C.1 relates model-implied and observed R&D to product-market and technology centrality and reports how the social-private return ratio varies with these measures.

As a separate external validation, we construct depreciated patent stocks from DISCERN-linked grants. The stocks depreciate at 15% per year and weight each patent either equally or by one plus the log of one plus its forward citations; within each year, we rescale them to the aggregate level of recovered knowledge capital. Log recovered knowledge capital has correlations of 0.65 and 0.64 with the count- and log-citation-weighted stocks, respectively; the corresponding mean annual correlations are 0.69 and 0.68.

4 Numerical Analysis

This section reports the quantitative implications of the estimated model. Aggregate R&D resource cost in the competitive equilibrium is below the R&D-only planner level, and effort is misallocated across firms. A uniform subsidy captures most of the R&D-only planner gain, whereas allowing product-market reallocation produces a much larger welfare effect.

4.1 Computation and Scale

Each product-market regime yields closed-form mappings from knowledge capital to quantities, output, and producer profit. Conditional on these mappings, we solve for R&D feedback rules and compute producer value and household welfare from continuous-time Lyapunov equations. Blockwise updates make the 757-firm competitive game tractable while retaining continuous states and both empirical networks; Sections C.3 and C.4 describes the algorithm and computational scale.

4.2 Social-Private R&D Wedges

We compare private and social marginal returns to R&D at the observed 2017 knowledge-capital vector. The private marginal return is the marginal increase in firm value induced by one additional unit of firm i's R&D effort before subtracting the marginal cost of that effort. Since one additional unit of x_i raises the drift of z_i by μ, this private marginal benefit is

$$\mathrm{PMR}_i(\boldsymbol{z}) \equiv \mu \frac{\partial V^i(\boldsymbol{z})}{\partial z_i} = 2\mu \left(\boldsymbol{X}_i^i\right)^\top \boldsymbol{z}.$$

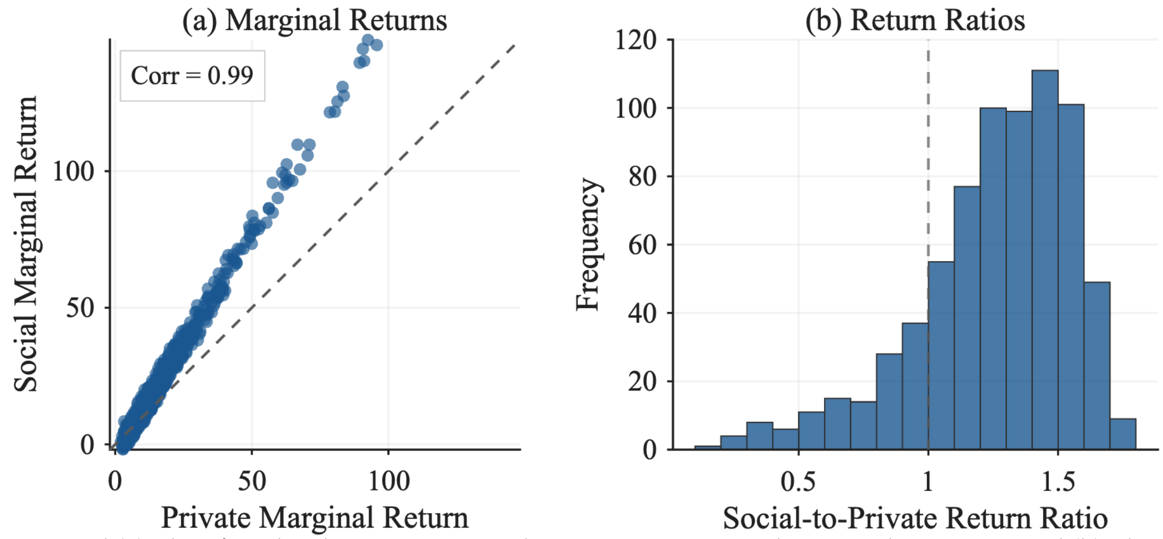
In vector form, $\mathrm{PMR}(\boldsymbol{z}) = 2\mu \widetilde{\boldsymbol{X}} \boldsymbol{z}$. Similarly, define the social marginal return as the increase in household welfare from one additional unit of firm i's R&D effort,

$$\mathrm{SMR}_i(\boldsymbol{z}) \equiv \mu \frac{\partial V(\boldsymbol{z})}{\partial z_i} = 2\mu \, \boldsymbol{e}_i^\top \boldsymbol{X} \boldsymbol{z}.$$

In vector form, $\mathrm{SMR}(\boldsymbol{z}) = 2\mu \boldsymbol{X} \boldsymbol{z}$. The social-private R&D wedge and the social-to-private return ratio are

$$\Delta_i(\boldsymbol{z}) \equiv \mathrm{SMR}_i(\boldsymbol{z}) - \mathrm{PMR}_i(\boldsymbol{z}), \qquad \mathcal{W}_i(\boldsymbol{z}) \equiv \frac{\mathrm{SMR}_i(\boldsymbol{z})}{\mathrm{PMR}_i(\boldsymbol{z})}.$$

Figure 3: Social-Private R&D Wedges: Marginal Returns and Return Ratios



Notes: Panel (a) plots firm-level private marginal returns against social marginal returns. Panel (b) plots the distribution of $\mathrm{SMR}_i(z)/\mathrm{PMR}_i(z)$ among firms with $\mathrm{PMR}_i(z) > 0$ and $\mathrm{SMR}_i(z) > 0$; the distribution trims the top and bottom 2 percent for readability, and the dashed vertical line marks equality of social and private returns. Returns are computed using the estimated networks and firm-level knowledge capital in 2017.

When both marginal returns are positive, we also report the local first-order-condition (FOC) subsidy measure

$$s_i^{\mathrm{loc}}(z) \equiv 1 - \frac{\mathrm{PMR}_i(z)}{\mathrm{SMR}_i(z)}.$$

We report $s_i^{\mathrm{loc}}(z)$ as a local summary of the sign and magnitude of the observed-state wedge.

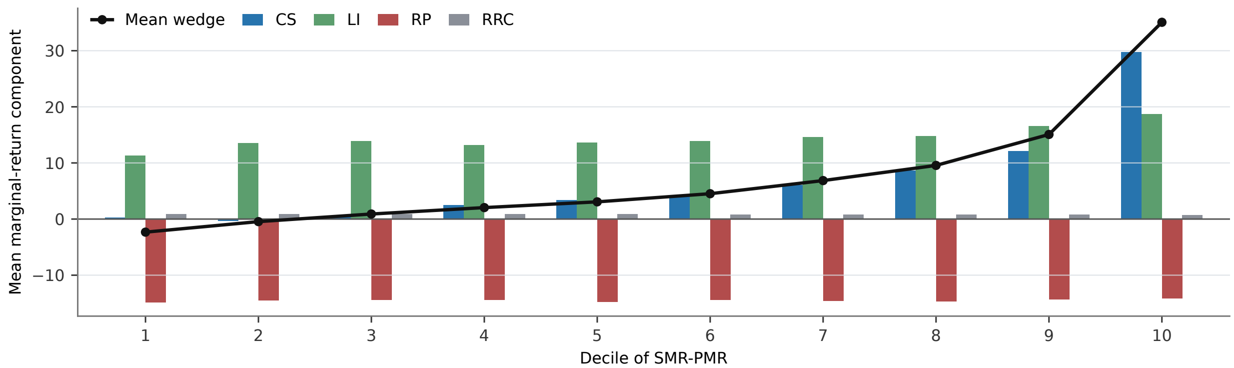
Figure 3 summarizes the firm-level wedge measures by plotting marginal-return levels and the social-to-private return-ratio distribution. The model-implied correlation between private and social marginal returns is high (0.994 after the figure's percentile trimming), but social returns tend to exceed private returns, especially among firms with high R&D returns.

Among the 739 firms with positive private and social marginal returns, the median social-to-private return ratio is 1.31, and social returns exceed private returns for 82.3% of firms. The distribution is heterogeneous: 22.5% of firms have ratios above 1.5, while the lower tail implies negative local FOC subsidies. The median local FOC subsidy is 23.4%.

The return-ratio distribution documents heterogeneity in the relative wedge, but the ratio does not directly admit an additive payoff decomposition. We therefore turn to the marginal-return difference, $\Delta_i(z) = \mathrm{SMR}_i(z) - \mathrm{PMR}_i(z)$. The exact decomposition is

$$\Delta_i(z) = \Delta_i^{\mathrm{CS}}(z) + \Delta_i^{\mathrm{LI}}(z) + \Delta_i^{\mathrm{RP}}(z) + \Delta_i^{\mathrm{RRC}}(z),$$

Figure 4: Wedge Components by SMR-PMR Decile



Notes: Firms are sorted into deciles by the additive marginal-return wedge, $\Delta_i(z) = \mathrm{SMR}_i(z) - \mathrm{PMR}_i(z)$, in the 739-firm positive marginal-return sample at the observed 2017 state. The black line reports the decile mean of the total wedge. Bars report decile means of the exact deterministic Lyapunov components defined in Section B.5. Units are model marginal-return units. CS, LI, RP, and RRC denote consumer-surplus, labor-income, rival-profit, and closed-loop rival R&D resource-cost components.

where CS and LI are the dynamic marginal values of consumer surplus and labor income generated by firm i's R&D. Together they measure the household value not reflected in aggregate producer gross profit—the appropriability gap. RP captures the effect on rivals' gross profits and is negative when innovation reallocates demand or profits away from rivals through business stealing. RRC captures the resulting change in rivals' discounted R&D resource costs under their competitive feedback policies. The four terms organize payoff incidence, while spillovers shape every component through equilibrium feedback. Under the sign convention used here, a positive component raises the social marginal return relative to the private marginal return.

The component averages show that consumer surplus and labor income generate the upward force in the wedge distribution, while rival-profit effects offset their levels and RRC is small. The mean wedge rises from −2.33 in the bottom decile to 35.07 in the top decile. Over the same deciles, CS rises from 0.28 to 29.73 and LI from 11.31 to 18.71. By comparison, RP remains between −14.80 and −14.08, and RRC lies between 0.70 and 0.88. CS and LI therefore account for most of the cross-decile variation in the additive social-private wedge.

4.3 Counterfactual Allocations

Table 5 compares the five allocations defined in Table 1 at the common 2017 knowledge-capital distribution. CC is the decentralized benchmark. CM and CS centralize R&D decisions while firms remain Cournot competitors; MM and SS centralize both production and R&D. The baseline instantaneous output-growth rate is 1.50%, comprising a 3.21

Table 5: Scenario Comparison in 2017

Scenario	Output (CC=100)	R&D resource cost (CC=100)	Instantaneous g_Y (%)	Welfare (CC=100)	Producer value share (%)
Baseline					
CC	100.00	100.00	1.50	100.00	27.08
R&D-only control, Cournot product markets					
CM	100.00	53.76	1.35	98.91	27.94
CS	100.00	231.02	1.78	100.50	25.83
Joint product-market and R&D control					
MM	96.33	62.63	1.39	95.66	43.50
SS	109.29	339.43	1.86	108.97	-5.83

Notes: All columns are evaluated at the observed 2017 knowledge-capital vector. Columns labeled CC=100 are normalized to the competitive equilibrium. R&D resource cost is $\sum_i x_i(z)^2$ under the policy used to compute growth and welfare. The instantaneous output-growth rate is $g_Y = d\log Y_t/dt$. Producer value share is $100 \times V^P/V$, net of R&D costs; its negative value in SS reflects discounted gross producer payoffs below discounted R&D costs. With linear utility, the consumption-equivalent welfare change for scenario k is $100 \times (V^k/V^{CC} - 1)$.

percentage-point contribution from technology spillovers, a 0.52 point contribution from direct R&D, and a 2.23 point drag from obsolescence. Appendix Sections B.6 and B.7 derives and decomposes g_Y.

4.3.1 Constrained R&D Control

CM and CS centralize R&D decisions while firms remain Cournot competitors. Given the same initial knowledge-capital distribution, Cournot conduct implies identical contemporaneous production quantities. Consequently, output remains $Y = 100$ across CC, CM, and CS at the observed state, although future quantities evolve with the scenario-specific knowledge-capital paths.

CM, interpretable as an economy-wide research joint venture that leaves Cournot conduct unchanged, values innovation through aggregate producer value and cuts R&D resource cost to 53.76. It internalizes effects on rival producers but not the household surplus that firms fail to appropriate. CS internalizes that household surplus and raises R&D resource cost to 231.02. Accordingly, g_Y falls to 1.35% under CM and rises to 1.78% under CS.

Because aggregate R&D resource cost is already below the constrained planner benchmark in the competitive equilibrium, the constrained monopolist's lower R&D effort pushes welfare down, leading to a 1.09% consumption-equivalent welfare loss relative to the competitive equilibrium outcome. The constrained planner, by contrast, lifts consumption-equivalent welfare 0.50% above competition. The producer value share moves in the opposite direction: prioritizing aggregate producer value raises the monopolist's share to 27.94%, whereas the constrained planner steers more surplus toward consumers and reduces the share to 25.83%. This constrained-planner result is the aggregate counterpart of the wedge measures above: many firms have social marginal R&D returns above private marginal returns, but the planner also reallocates effort across firms according to these heterogeneous gaps.

4.3.2 Joint Product-Market and R&D Control

In MM and SS, the monopolist or planner chooses both production and dynamic R&D effort. Comparing these allocations with CC shows the joint effect of centralizing production and innovation; CM and CS isolate the R&D-only component.

MM reduces output by 3.67% relative to competition, whereas the full planner raises output by 9.29%. These product-market changes feed directly into R&D. MM reduces R&D resource cost to 62.63, above CM's 53.76 because monopoly pricing raises private returns to innovation. Under SS, R&D resource cost rises to 339.43, up from 231.02 under CS and equal to 3.39 times its competitive level, because the planner's product-market expansion raises the social return to innovation. This magnitude lies within the estimate of Jones and Williams (1998) that optimal R&D spending is at least two to four times actual spending; see also Jones and Williams (2000). The direction of the result is also consistent with the finding of Bloom et al. (2013) that technology spillovers dominate product-market business stealing, although their approximately 3.5-to-one estimate concerns social relative to private returns rather than the optimal level of R&D spending.

These joint changes in production and R&D effort affect instantaneous g_Y. Under MM, g_Y is 1.39%, 0.11 percentage points below competition but 0.04 points above CM. Under SS, g_Y rises to 1.86%, 0.36 points above competition and 0.08 points above CS. The planner's reallocation accelerates knowledge accumulation, whereas the monopolist's restraint keeps growth below competition.

Joint product-market and R&D control has much larger welfare effects than R&D-only control. MM lowers consumption-equivalent welfare by 4.34% relative to competition, compared with a 1.09% loss under CM. SS raises welfare by 8.97%, compared with a 0.50% gain under CS. These differences reflect both the immediate output effects of product-market reallocation and the resulting changes in returns to R&D.

The full-planner welfare gain is of the same order of magnitude as the static oligopoly loss estimated with the same GHL demand structure. Pellegrino (2025) estimates that oligopoly lowers total surplus by about 11.5% for publicly traded corporations. In our 2017 patenting-firm sample, SS raises welfare by 8.97% and contemporaneous output by

9.29%. Our calculation also allows R&D to respond endogenously and uses the linked Compustat–DISCERN sample.

4.4 Uniform R&D Subsidy

As summarized in Table 1, a uniform R&D subsidy reduces each firm's private R&D cost from $x_{i,t}^2$ to $(1-s)x_{i,t}^2$, where $s \in [0,1)$ is the subsidy rate. The subsidy changes firms' dynamic R&D incentives while product-market conduct remains Cournot, $\boldsymbol{q} = \boldsymbol{N}\boldsymbol{z}$. Given a value-function coefficient matrix, the feedback rule becomes

$$\boldsymbol{x}_t = \frac{1}{1-s}\mu\widetilde{\boldsymbol{X}}(s)\boldsymbol{z}_t.$$

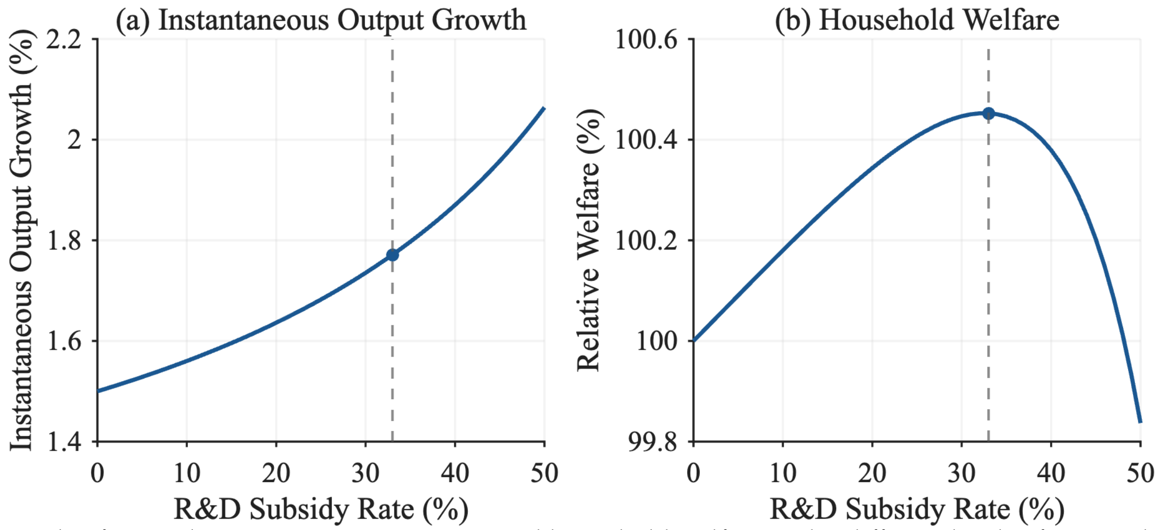
We treat subsidy payments as financed by lump-sum taxes, so the subsidy changes private incentives but not the aggregate resource cost of R&D; household welfare therefore continues to subtract the full resource cost $\boldsymbol{x}_t^T\boldsymbol{x}_t$. Appendix Section B.4 records the subsidy-specific Riccati and welfare equations. At the 2017 state, welfare is maximized at a 33% subsidy on the $s = 0, 0.01, \ldots, 0.50$ grid (Figure 5). The subsidy raises g_Y to 1.77% and welfare by 0.45%, nearly matching the constrained planner's 1.78% growth rate and 0.50% welfare gain. Beyond 33%, additional R&D costs outweigh the growth benefits. Appendix Figure 7 reports the transition paths, which overtake the competitive consumption path after about 9.5 years.

4.4.1 Scale versus Allocation

How much of the constrained planner's gain comes from expanding the aggregate scale of R&D, and how much from reallocating it across firms? Both allocations use linear feedback rules $\boldsymbol{x} = \boldsymbol{K}\boldsymbol{z}$. We measure scale by resource cost $R(\boldsymbol{K}) = \|\boldsymbol{K}\boldsymbol{z}_0\|^2$ and hold allocation shape fixed by proportional rescaling. With $\lambda = \sqrt{R(\boldsymbol{K}_S)/R(\boldsymbol{K}_C)} = 1.52$, $\lambda\boldsymbol{K}_C$ combines planner scale with the competitive allocation, while $\lambda^{-1}\boldsymbol{K}_S$ combines competitive scale with the planner allocation. Table 6 evaluates these scale-allocation combinations.

Scale dominates. Expanding aggregate scale alone raises consumption-equivalent welfare by 0.446%, recovering about nine-tenths of the constrained planner's 0.50% gain, whereas imposing the planner's allocation at competitive scale raises welfare by only 0.020%. The 33% uniform subsidy moves R&D resource cost to 230.91, almost exactly the CS level of 231.02, and delivers a 0.452% welfare gain. This similarity in aggregate R&D scale helps explain why the subsidy nearly matches CS welfare. Appendix Section B.4 reports the order-based decomposition and firm-level allocation comparison.

Figure 5: Instantaneous 2017 Output Growth and Household Welfare under R&D Subsidy



Notes: This figure plots instantaneous 2017 g_Y and household welfare under different levels of R&D subsidy. The dashed vertical line and filled marker identify the welfare-maximizing 33% subsidy. The product-market rivalry and technology-spillover networks, as well as the distribution of knowledge capital, are estimated using data from 2017.

Table 6: Scale versus Allocation in the Constrained-Planner R&D Gain

Rule	Aggregate scale	Allocation shape	R&D resource cost (CC=100)	Instantaneous g_Y (%)	Welfare gain (%)
CC	CC	CC	100.00	1.50	0.000
Scale only	CS	CC	231.02	1.77	0.446
Allocation only	CC	CS	100.00	1.51	0.020
33% uniform subsidy	—	—	230.91	1.77	0.452
CS	CS	CS	231.02	1.78	0.498

Notes: All rows are evaluated at the observed 2017 knowledge-capital vector with Cournot product-market conduct, $\boldsymbol{q} = \boldsymbol{N}\boldsymbol{z}$, and the R&D effort rule is the linear feedback $\boldsymbol{x} = \boldsymbol{K}\boldsymbol{z}$. Aggregate scale is the R&D resource cost $R(\boldsymbol{K}) = \|\boldsymbol{K}\boldsymbol{z}_0\|^2$. Allocation shape is the normalized feedback rule: with $\lambda = \sqrt{R(\boldsymbol{K}_S)/R(\boldsymbol{K}_C)} = 1.520$, the "scale only" rule $\lambda\boldsymbol{K}_C$ holds the competitive allocation shape and moves aggregate scale to the CS level, while the "allocation only" rule $\lambda^{-1}\boldsymbol{K}_S$ holds competitive scale and adopts the planner shape. Welfare gain is $100 \times (\boldsymbol{z}_0^\top \boldsymbol{X}\boldsymbol{z}_0/\boldsymbol{z}_0^\top \boldsymbol{X}^{CC}\boldsymbol{z}_0 - 1)$. The 33% uniform subsidy is the grid-optimal implementable policy shown for reference.

4.5 Robustness

The main scenario comparison uses $\beta = 0.02$, $\alpha = 0.12$, and $\rho = 10\%$. Appendix Sections B.10 to B.12 and Tables 11 to 13 report recalibrated sensitivity exercises. Across the reported values of β and α, the R&D-only planner gain ranges from 0.37% to 0.76%, and the grid-optimal uniform subsidy ranges from 29% to 37%. Lower discount rates raise both the R&D-only gain and the optimal subsidy rate but do not change the scenario ranking.

In every case, the full-planner gain is substantially larger.

5 Conclusion

This paper develops and estimates an endogenous growth model in which many oligopolistic firms interact through empirically measured product-market rivalry and technology-spillover networks. The linear-quadratic structure reduces the dynamic R&D game to a coupled system of Riccati equations solvable at the scale of the observed firm distribution.

At the observed 2017 state, competitive R&D is below the constrained-planner level, and heterogeneous social-private wedges generate firm-level misallocation. Despite this heterogeneity, a 33% uniform R&D subsidy delivers a 0.45% consumption-equivalent welfare gain, close to the constrained planner's 0.50% gain, because expanding the aggregate scale of R&D accounts for most of the improvement. The constrained monopolist, which centralizes R&D decisions while retaining Cournot product-market conduct, instead lowers both R&D and welfare. A full planner that jointly reallocates product-market quantities and R&D raises welfare by 8.97%.

These results show that a uniform subsidy captures most of the welfare gains available through R&D policy alone, while the much larger full-planner gain highlights the quantitative importance of product-market distortions and their interaction with innovation incentives. The full-planner welfare gain is of the same order of magnitude as static oligopoly losses estimated using the same demand structure, underscoring the limits of evaluating innovation policy separately from product-market allocation. Because the analysis takes the set of firms and the two networks as given, incorporating entry and endogenous network formation is a natural extension.

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Appendix

Appendix A Model Derivations

A.1 GHL Demand Reduction

The GHL demand system begins with common characteristics and idiosyncratic product characteristics. Product i is described by an m-dimensional nonnegative vector $\boldsymbol{\psi}_i$ of common characteristics, normalized so that $\boldsymbol{\psi}_i^T\boldsymbol{\psi}_i = 1$. Stacking these vectors gives

$$\boldsymbol{\Psi} \equiv \begin{bmatrix} \boldsymbol{\psi}_1 & \boldsymbol{\psi}_2 & \cdots & \boldsymbol{\psi}_n \end{bmatrix},$$

and the product-similarity matrix is

$$\boldsymbol{S} \equiv \boldsymbol{\Psi}^T\boldsymbol{\Psi}.$$

The normalization implies $S_{ii} = 1$, while for $i \neq j$, $S_{ij} = \boldsymbol{\psi}_i^T\boldsymbol{\psi}_j$ is the cosine similarity between products' common-characteristic vectors.

Let $\boldsymbol{q}_t$ be the vector of product quantities. Common and idiosyncratic characteristic quantities are

$$\boldsymbol{y}_t^C = \boldsymbol{\Psi}\boldsymbol{q}_t \tag{17}$$

and

$$\boldsymbol{y}_t^I = \boldsymbol{q}_t. \tag{18}$$

The final-good producer aggregates these characteristics according to

$$Y_t = \alpha \sum_{k=1}^{m} \left(b_{k,t}^C y_{k,t}^C - \frac{1}{2}\left(y_{k,t}^C\right)^2 \right) + (1-\alpha) \sum_{i=1}^{n} \left(b_{i,t}^I y_{i,t}^I - \frac{1}{2}\left(y_{i,t}^I\right)^2 \right), \tag{19}$$

where $\alpha \in [0,1]$ is the weight on common characteristics. Substituting Equations (17) and (18) into Equation (19) gives

$$Y_t = \boldsymbol{q}_t^T \left[\alpha\boldsymbol{\Psi}^T\boldsymbol{b}_t^C + (1-\alpha)\boldsymbol{b}_t^I\right] - \frac{1}{2}\boldsymbol{q}_t^T \left[\alpha\boldsymbol{\Psi}^T\boldsymbol{\Psi} + (1-\alpha)\boldsymbol{I}\right]\boldsymbol{q}_t.$$

Thus the reduced-form product-quality vector and rivalry matrix are

$$\boldsymbol{b}_t \equiv \alpha\boldsymbol{\Psi}^T\boldsymbol{b}_t^C + (1-\alpha)\boldsymbol{b}_t^I, \qquad \boldsymbol{\Sigma} \equiv \alpha\boldsymbol{S} + (1-\alpha)\boldsymbol{I},$$

which yields the reduced-form aggregator in Equation (1).

A.2 Competitive Static Equilibrium

Each firm's real gross profit before subtracting R&D cost is

$$\begin{aligned}\pi_{i,t} &= \frac{p_{i,t}}{P_t} q_{i,t} - \frac{w_t}{P_t} l_{i,t} \\ &= \left(b_{i,t} - \frac{1}{a_{i,t}} \frac{w_t}{P_t} - \sum_j \sigma_{ij} q_{j,t} \right) q_{i,t},\end{aligned}$$

where the second line uses the inverse demand in Equation (6) and the production function in Equation (2). Firm i chooses $q_{i,t}$, $a_{i,t}$, and $b_{i,t}$, taking rivals' quantities, the real wage, and its own knowledge capital as given. The FOC with respect to quantity is

$$0 = b_{i,t} - \frac{1}{a_{i,t}} \frac{w_t}{P_t} - 2q_{i,t} - \sum_{j \neq i} \sigma_{ij} q_{j,t}. \tag{20}$$

The FOCs with respect to $a_{i,t}$ and $b_{i,t}$ imply

$$a_{i,t} = \sqrt{\frac{w_t}{\zeta P_t}} \tag{21}$$

$$b_{i,t} = z_{i,t} - \sqrt{\frac{\zeta w_t}{P_t}}. \tag{22}$$

Inserting Equation (21) into the labor-market-clearing condition in Equation (5) gives

$$\sqrt{\frac{\zeta w_t}{P_t}} = \frac{\zeta}{L} \sum_i q_{i,t}. \tag{23}$$

Inserting the optimality condition for technology choice Equation (21) and Equation (22) into the optimality condition for quantity choice Equation (20), and using $\sigma_{ii} = 1$ to write $2q_{i,t} + \sum_{j \neq i} \sigma_{ij} q_{j,t} = q_{i,t} + \sum_j \sigma_{ij} q_{j,t}$, we obtain

$$z_{i,t} = 2\sqrt{\zeta \frac{w_t}{P_t}} + q_{i,t} + \sum_j \sigma_{ij} q_{j,t}$$

Inserting the expression for real wage Equation (23), we obtain

$$z_{i,t} = q_{i,t} + \sum_j \left(2\frac{\zeta}{L} + \sigma_{ij}\right) q_{j,t}.$$

In vector form,

$$\boldsymbol{z}_t = \left(2\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{I}\right) \boldsymbol{q}_t.$$

Therefore, quantities are linear in knowledge capital:

$$\boldsymbol{q}_t = \boldsymbol{N}\boldsymbol{z}_t$$

where

$$\boldsymbol{N} \equiv \left(2\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{I}\right)^{-1}.$$

Lemma 1 (Static Cournot uniqueness)**.** *Suppose $\alpha \in [0,1]$ and $\zeta/L \geq 0$. Then $2\zeta\boldsymbol{J}/L + \boldsymbol{\Sigma} + \boldsymbol{I}$ is positive definite. Hence the interior static Cournot first-order conditions have a unique solution, given by $\boldsymbol{q}_t = \boldsymbol{N}\boldsymbol{z}_t$.*[2]

Proof. The GHL similarity matrix is $\boldsymbol{S} = \boldsymbol{\Psi}^T\boldsymbol{\Psi}$, so it is positive semidefinite. Since $\boldsymbol{\Sigma} = \alpha\boldsymbol{S} + (1-\alpha)\boldsymbol{I}$ and $\alpha \in [0,1]$, $\boldsymbol{\Sigma}$ is positive semidefinite. The all-ones matrix $\boldsymbol{J}$ is also positive semidefinite. Therefore, for any nonzero vector $\boldsymbol{v}$,

$$\boldsymbol{v}^T\left(2\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{I}\right)\boldsymbol{v} = 2\frac{\zeta}{L}\boldsymbol{v}^T\boldsymbol{J}\boldsymbol{v} + \boldsymbol{v}^T\boldsymbol{\Sigma}\boldsymbol{v} + \|\boldsymbol{v}\|^2 > 0.$$

The matrix multiplying $\boldsymbol{q}_t$ in the static first-order conditions is therefore nonsingular, so the linear system has exactly one interior solution. □

Combining the quantity FOC with $\sigma_{ii} = 1$ gives

$$b_{i,t} - \frac{1}{a_{i,t}}\frac{w_t}{P_t} - \sum_j \sigma_{ij} q_{j,t} = q_{i,t},$$

so the equilibrium static gross profit is $\pi_{i,t} = q_{i,t}^2$.

[2] If nonnegativity of quantities were imposed in the static block, the same positive-definite matrix is a P-matrix, so the associated linear-complementarity problem also has a unique solution for each $\boldsymbol{z}_t$ (Cottle et al., 1992).

A.3 Output in Competitive Equilibrium

Let $\mathbf{1}$ denote an $n \times 1$ vector with all elements equal to 1. Rewrite Equation (22) in vector form:

$$\boldsymbol{b}_t = \boldsymbol{z}_t - \sqrt{\zeta \frac{w_t}{P_t}} \mathbf{1}$$

Inserting Equation (23), we obtain

$$\begin{aligned}
\boldsymbol{b}_t &= \boldsymbol{z}_t - \frac{\zeta}{L} \left(\sum_i q_{i,t} \right) \mathbf{1} \\
&= \boldsymbol{z}_t - \frac{\zeta}{L} \boldsymbol{J} \boldsymbol{q}_t.
\end{aligned}$$

From Equation (7), we obtain

$$\boldsymbol{b}_t = \left(\frac{\zeta}{L} \boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{I} \right) \boldsymbol{q}_t$$

Therefore, from Equation (1), the total output is given by

$$\begin{aligned}
Y_t &= \boldsymbol{q}_t^T \left(\frac{\zeta}{L} \boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{I} \right) \boldsymbol{q}_t - \frac{1}{2} \boldsymbol{q}_t^T \boldsymbol{\Sigma} \boldsymbol{q}_t \\
&= \boldsymbol{q}_t^T \left(\frac{\zeta}{L} \boldsymbol{J} + \frac{1}{2} \boldsymbol{\Sigma} + \boldsymbol{I} \right) \boldsymbol{q}_t \\
&= \boldsymbol{z}_t^T \boldsymbol{N}^T \left(\frac{\zeta}{L} \boldsymbol{J} + \frac{1}{2} \boldsymbol{\Sigma} + \boldsymbol{I} \right) \boldsymbol{N} \boldsymbol{z}_t \\
&= \boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t
\end{aligned}$$

where

$$\boldsymbol{Q} \equiv \boldsymbol{N}^T \left(\frac{\zeta}{L} \boldsymbol{J} + \frac{1}{2} \boldsymbol{\Sigma} + \boldsymbol{I} \right) \boldsymbol{N}$$

A.4 Competitive Dynamic R&D Derivation

For the quadratic value-function guess in Equation (9), with $\boldsymbol{X}^i$ symmetric, the first and second derivatives are

$$\frac{\partial V^i(\boldsymbol{z})}{\partial \boldsymbol{z}} = 2 \left(\boldsymbol{X}^i \boldsymbol{z} \right)^T \tag{24}$$

$$\frac{\partial^2 V^i(\boldsymbol{z})}{\partial \boldsymbol{z} \partial \boldsymbol{z}^T} = 2 \boldsymbol{X}^i. \tag{25}$$

The derivative of Equation (8) with respect to x_i is $-2x_i + \mu \left[V_z^i(z)\right]_i = 0$. Since $\left[V_z^i(z)\right]_i = 2\left(\boldsymbol{X}_i^i\right)^T \boldsymbol{z}$, the first-order condition yields the linear policy in Equation (10).

The diffusion term in Equation (8) simplifies to

$$\frac{\gamma^2}{2} \operatorname{Tr}\left[\operatorname{diag}(\boldsymbol{z})\, 2\boldsymbol{X}^i \operatorname{diag}(\boldsymbol{z})\right] = \gamma^2 \boldsymbol{z}^T \mathcal{D}\left(\boldsymbol{X}^i\right) \boldsymbol{z}.$$

Substituting Equations (9), (10), (24) and (25) into Equation (8) and collecting the quadratic terms in $\boldsymbol{z}$ gives the stacked Riccati system in Equation (11).

Proof of Theorem 1. With rivals' feedback rules fixed, firm i faces a single-agent linear-quadratic control problem. Let $V^i(\boldsymbol{z}) = \boldsymbol{z}^T \boldsymbol{X}^i \boldsymbol{z}$. The Riccati equation implies that V^i satisfies firm i's HJB equation. The maximand in the HJB is strictly concave in x_i because the flow cost contains $-x_i^2$, and its first-order condition gives the feedback rule in Equation (10). For any admissible control, Itô's formula applied to $e^{-\rho t} V^i(\boldsymbol{z}_t)$ on $[0, T]$ and the HJB inequality imply that the expected discounted payoff is no larger than

$$V^i(\boldsymbol{z}_0) - \mathbb{E}_0\left[e^{-\rho T} V^i(\boldsymbol{z}_T)\right],$$

with equality under the feedback rule above. Taking $T \to \infty$, using the admissibility condition for arbitrary controls, and using Definition 2 for the candidate feedback rule gives optimality of the candidate response. Since this argument applies to every firm, the stabilizing Riccati solution is a Markov perfect equilibrium in the stated admissible class. □

A.5 Competitive Welfare and Producer Value

After firms' equilibrium policies are fixed, household welfare is obtained from the valuation equation

$$\rho V(\boldsymbol{z}) = \boldsymbol{z}^T \boldsymbol{Q} \boldsymbol{z} - \mu^2 \boldsymbol{z}^T \widetilde{\boldsymbol{X}}^T \widetilde{\boldsymbol{X}} \boldsymbol{z} + V_z(\boldsymbol{z}) \boldsymbol{\Phi} \boldsymbol{z} + \frac{\gamma^2}{2} \operatorname{Tr}\left[\operatorname{diag}(\boldsymbol{z})\, V_{zz}(\boldsymbol{z}) \operatorname{diag}(\boldsymbol{z})\right].$$

The first two terms are household consumption, $Y_t - \sum_i x_{i,t}^2$, and the last two terms capture expected changes in continuation value. With $V(\boldsymbol{z}) = \boldsymbol{z}^T \boldsymbol{X} \boldsymbol{z}$, substituting the derivatives from Equations (24) and (25) with $\boldsymbol{X}^i$ replaced by $\boldsymbol{X}$ gives the Lyapunov equation in Equation (12).

Table 7: Growth Rate of Variables in Deterministic Economy

Growth rate	Variables
0	$l_{i,t}$
g	$a_{i,t}$, $b_{i,t}$, $z_{i,t}$, $q_{i,t}$, $x_{i,t}$, $p_{i,t}/P_t$
$2g$	$\pi_{i,t}$, C_t, Y_t, w_t/P_t

Notes: Growth rates are continuous-time instantaneous rates evaluated along the balanced-growth-path equilibrium in the deterministic economy with no shocks.

Aggregate producer value in the competitive equilibrium is

$$V^P\left(\boldsymbol{z}_0\right) = \mathbb{E}_0\left[\int_0^\infty \exp\left(-\rho t\right)\left(\sum_i \pi_{i,t} - \sum_i x_{i,t}^2\right) dt \middle| \boldsymbol{z}_0\right].$$

Because aggregate real profit is $\sum_i \pi_{i,t} = \boldsymbol{z}_t^T \boldsymbol{N}^T \boldsymbol{N} \boldsymbol{z}_t$, the producer-value coefficient $\boldsymbol{X}^P$ solves the same Lyapunov equation as Equation (12), replacing $\boldsymbol{Q}$ with $\boldsymbol{N}^T\boldsymbol{N}$.

A.6 Deterministic Balanced Growth Path

We characterize the existence of a balanced growth path in the deterministic economy, $\gamma = 0$, using the closed-loop system $\dot{\boldsymbol{z}}_t = \boldsymbol{\Phi}\boldsymbol{z}_t$. When $\gamma > 0$, the same transition matrix governs the drift of knowledge capital. Table 7 summarizes the implied growth rates of the main variables along a deterministic balanced growth path.

We impose the following spectral condition on $\boldsymbol{\Phi}$, which is sufficient for the existence of a balanced-growth-path equilibrium.

Assumption 2. *The deterministic closed-loop technology transition matrix $\boldsymbol{\Phi}$ has a simple real dominant eigenvalue $g > 0$ with an associated strictly positive right eigenvector, and all other eigenvalues have real parts strictly below g.*

This assumption imposes the spectral property needed for balanced growth: the closed-loop technology system has a unique positive long-run growth direction. It rules out cases with multiple dominant growth directions or a non-positive balanced-growth-path knowledge-capital distribution.

Definition 3. Balanced Growth Path Equilibrium: A *balanced-growth-path equilibrium* is a linear Markov perfect equilibrium such that all firms' knowledge capital grows at the same constant rate.

Proposition 1. *Consider the deterministic economy, $\gamma = 0$. Let Assumptions 1 and 2 hold, and let $\bar{\boldsymbol{z}} \gg 0$ be the right eigenvector of $\boldsymbol{\Phi}$ associated with the dominant eigenvalue g. Then, for any scalar $c > 0$, there exists a balanced-growth-path equilibrium whose knowledge-capital path is*

$$\boldsymbol{z}_t = e^{gt} c \bar{\boldsymbol{z}}.$$

Along this path, all firms' knowledge capital grows at rate g, and the cross-sectional distribution of knowledge capital is proportional to $\bar{\boldsymbol{z}}$.

Proof. Assumption 1 gives a stabilizing solution to the stacked Riccati equations and hence linear Markov R&D policies, together with the static equilibrium mappings characterized above. In the deterministic economy, these policies imply the closed-loop system $\dot{\boldsymbol{z}}_t = \boldsymbol{\Phi} \boldsymbol{z}_t$. Since $\boldsymbol{\Phi}\bar{\boldsymbol{z}} = g\bar{\boldsymbol{z}}$, the path $\boldsymbol{z}_t = e^{gt} c \bar{\boldsymbol{z}}$ satisfies this system. Because $\bar{\boldsymbol{z}} \gg 0$ and $c > 0$, all firms' knowledge capital remains positive and grows at the common rate g. The induced static allocations and R&D policies therefore constitute a linear Markov perfect equilibrium on a balanced growth path. □

Proposition 1 implies that the equilibrium growth rate of knowledge capital is the dominant eigenvalue of $\boldsymbol{\Phi}$, and the balanced-growth-path knowledge-capital distribution is the associated eigenvector. Since $\boldsymbol{q}_t = \boldsymbol{N} \boldsymbol{z}_t$ and equilibrium R&D strategies are linear in $\boldsymbol{z}_t$, quantities and R&D grow at rate g. The static optimality conditions imply that product quality, labor productivity, and real product prices also grow at rate g, while the real wage, output, consumption, and profits grow at rate $2g$ because they are pinned down by quadratic terms in knowledge capital. In the deterministic case, Assumption 1 requires $\rho > 2g$, so discounted payoffs are finite along the balanced growth path.

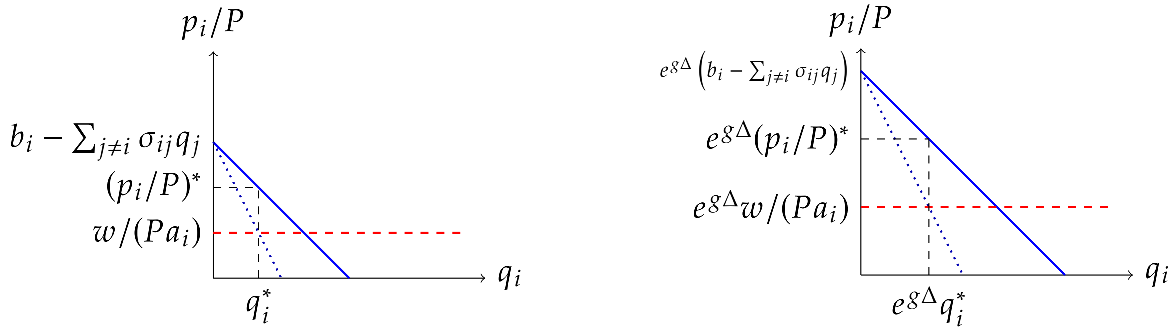
Figure 6 illustrates the mechanics of a balanced-growth-path equilibrium in the deterministic economy. In equilibrium, quality, quantity, real price, and real marginal cost grow at the same continuous-time rate g, so the partial-equilibrium diagram expands homothetically along both axes.

Appendix B Counterfactual and Wedge Derivations

B.1 Social Planner's Static Problem

We reduce the planner's static problem by substituting the knowledge-allocation constraint $b_{i,t} = z_{i,t} - \zeta a_{i,t}$ and the production technology $l_{i,t} = q_{i,t}/a_{i,t}$ into the planner's objective

Figure 6: Partial Equilibrium Diagram and Its Proportional Growth



Notes: The left panel depicts the static equilibrium for a firm's product, showing the residual demand (blue), marginal revenue (dotted line), and real marginal cost (red) curves. The right panel illustrates how the real equilibrium price, quantity, real marginal cost, and the intercept of the residual demand scale proportionally with the growth factor $\exp(g\Delta)$ over an interval of length Δ.

and labor-market constraint. Define the Lagrangian as

$$\mathcal{L} = \boldsymbol{q}_t^T (\boldsymbol{z}_t - \zeta \boldsymbol{a}_t) - \frac{1}{2}\boldsymbol{q}_t^T \boldsymbol{\Sigma} \boldsymbol{q}_t - \lambda_t \left\{ \sum_i \frac{q_{i,t}}{a_{i,t}} - L \right\}$$

We focus on the solution in which the labor constraint binds. The FOC for $a_{i,t}$ implies $\lambda_t = \zeta a_{i,t}^2 > 0$, so the square root below is well defined. The FOC w.r.t. $a_{i,t}$ gives

$$a_{i,t} = \sqrt{\frac{\lambda_t}{\zeta}} \tag{26}$$

and therefore,

$$b_{i,t} = z_{i,t} - \sqrt{\zeta \lambda_t} \tag{27}$$

Inserting Equation (26) into Equation (13), we obtain

$$\sqrt{\zeta \lambda_t} = \frac{\zeta}{L} \sum_i q_{i,t}$$

Note that

$$\sqrt{\zeta \lambda_t} \mathbf{1} = \frac{\zeta}{L} \left(\sum_i q_{i,t} \right) \mathbf{1} = \frac{\zeta}{L} \boldsymbol{J} \boldsymbol{q}_t. \tag{28}$$

The FOC w.r.t. $\boldsymbol{q}_t$ gives

$$\boldsymbol{z}_t - \zeta \boldsymbol{a}_t - \boldsymbol{\Sigma} \boldsymbol{q}_t = \lambda_t \begin{bmatrix} 1/a_{1,t} \\ 1/a_{2,t} \\ \vdots \\ 1/a_{n,t} \end{bmatrix}$$

Substituting Equation (26), we obtain

$$\boldsymbol{z}_t = 2\sqrt{\zeta \lambda_t}\mathbf{1} + \boldsymbol{\Sigma} \boldsymbol{q}_t$$

Using Equation (28), we obtain

$$\boldsymbol{z}_t = \left(2\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma}\right) \boldsymbol{q}_t$$

Therefore,

$$\boldsymbol{q}_t = \boldsymbol{N}^* \boldsymbol{z}_t \tag{29}$$

where

$$\boldsymbol{N}^* = \left(2\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma}\right)^{-1}$$

Because $\boldsymbol{J}$ and $\boldsymbol{\Sigma}$ are symmetric, $\boldsymbol{N}^*$ is symmetric as well. To obtain the quadratic form of output, substitute Equation (27) into Equation (1):

$$Y_t = \boldsymbol{q}_t^T \left(\boldsymbol{z}_t - \sqrt{\zeta \lambda_t}\mathbf{1}\right) - \frac{1}{2}\boldsymbol{q}_t^T \boldsymbol{\Sigma} \boldsymbol{q}_t$$

Using Equation (28) together with Equation (29),

$$\begin{aligned} Y_t &= \boldsymbol{q}_t^T \left(\left(2\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma}\right) \boldsymbol{q}_t - \frac{\zeta}{L}\boldsymbol{J}\boldsymbol{q}_t\right) - \frac{1}{2}\boldsymbol{q}_t^T \boldsymbol{\Sigma} \boldsymbol{q}_t \\ &= \boldsymbol{q}_t^T \left(\frac{\zeta}{L}\boldsymbol{J} + \frac{1}{2}\boldsymbol{\Sigma}\right) \boldsymbol{q}_t \\ &= \frac{1}{2}\boldsymbol{q}_t^T (\boldsymbol{N}^*)^{-1} \boldsymbol{q}_t \\ &= \frac{1}{2}\boldsymbol{z}_t^T \boldsymbol{N}^* \boldsymbol{z}_t \end{aligned}$$

B.2 Planner Dynamic R&D Derivation

The HJB equation for the social planner's dynamic problem is

$$\rho V^{SS}(\boldsymbol{z}) = \max_{\boldsymbol{x}} \left\{ \frac{1}{2}\boldsymbol{z}^T \boldsymbol{N}^* \boldsymbol{z} - \boldsymbol{x}^T \boldsymbol{x} + V_z^{SS}(\boldsymbol{z}) \left[(\boldsymbol{\Omega} - \delta \boldsymbol{I}) \boldsymbol{z} + \mu \boldsymbol{x} \right] \right.$$
$$\left. + \frac{\gamma^2}{2} \mathrm{Tr} \left[\mathrm{diag}(\boldsymbol{z}) V_{zz}^{SS}(\boldsymbol{z}) \mathrm{diag}(\boldsymbol{z}) \right] \right\}. \tag{30}$$

Using the quadratic guess

$$V^{SS}(\boldsymbol{z}) = \boldsymbol{z}^T \boldsymbol{X}^{SS} \boldsymbol{z}, \tag{31}$$

the first-order condition with respect to $\boldsymbol{x}$ is $-2\boldsymbol{x} + \mu \left(V_z^{SS}(\boldsymbol{z}) \right)^T = 0$, which gives the linear R&D rule in Equation (14). The induced transition is $\boldsymbol{\Phi}^{SS} = \boldsymbol{\Omega} + \mu^2 \boldsymbol{X}^{SS} - \delta \boldsymbol{I}$. We take the value-function coefficient matrix to be symmetric, since only its symmetric part affects $\boldsymbol{z}^T \boldsymbol{X}^{SS} \boldsymbol{z}$. Thus the drift term is represented by $\boldsymbol{X}^{SS} \boldsymbol{\Phi}^{SS} + (\boldsymbol{\Phi}^{SS})^T \boldsymbol{X}^{SS}$ in quadratic form. This path-flow convention matters because the spillover matrix $\boldsymbol{\Omega}$ is generally asymmetric. Substituting the quadratic value function, the linear R&D rule, and this transition into Equation (30) gives the planner Riccati equation in Equation (15).

B.3 Monopoly Allocation and Welfare

This subsection derives the full monopoly allocation (MM) used in the counterfactual analysis. We keep the superscript M for static monopoly matrices, such as $\boldsymbol{N}^M$, $\boldsymbol{P}^M$, and $\boldsymbol{Q}^M$, and use the superscript MM for the full scenario in which the monopolist controls both production and R&D. Since one decision maker controls the full R&D vector, the monopoly R&D policy is obtained from a single Riccati equation; household welfare under that policy is then computed separately from a Lyapunov equation.

Static Mapping

The monopolist chooses $\boldsymbol{a}_t$, $\boldsymbol{b}_t$, and $\boldsymbol{q}_t$ to maximize aggregate real profit. Differentiating Π_t with respect to $q_{i,t}$, and using the symmetry of $\boldsymbol{\Sigma}$ so that $\partial(\boldsymbol{q}_t^T \boldsymbol{\Sigma} \boldsymbol{q}_t)/\partial q_{i,t} = 2 \sum_j \sigma_{ij} q_{j,t}$, gives

$$0 = b_{i,t} - \frac{1}{a_{i,t}} \frac{w_t}{P_t} - 2 \sum_j \sigma_{ij} q_{j,t}. \tag{32}$$

The real profit of each product is therefore

$$
\begin{aligned}
\pi_{i,t} &= \left(b_{i,t} - \frac{1}{a_{i,t}}\frac{w_t}{P_t}\right) q_{i,t} - \sum_j \sigma_{ij} q_{i,t} q_{j,t} \\
&= 2\sum_j \sigma_{ij} q_{j,t} q_{i,t} - \sum_j \sigma_{ij} q_{i,t} q_{j,t} \\
&= \sum_j \sigma_{ij} q_{i,t} q_{j,t}. \qquad (33)
\end{aligned}
$$

The FOC with respect to $a_{i,t}$ gives

$$
a_{i,t} = \sqrt{\frac{1}{\zeta}\frac{w_t}{P_t}}, \qquad (34)
$$

and therefore,

$$
b_{i,t} = z_{i,t} - \sqrt{\zeta \frac{w_t}{P_t}}. \qquad (35)
$$

Inserting Equation (34) into the labor-market-clearing condition Equation (5) gives

$$
\sqrt{\zeta \frac{w_t}{P_t}} = \frac{\zeta}{L} \sum_i q_{i,t}. \qquad (36)
$$

Combining Equations (32), (35) and (36) yields

$$
\boldsymbol{z}_t = 2\left(\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma}\right)\boldsymbol{q}_t, \qquad (37)
$$

so monopoly quantities are linear in knowledge capital:

$$
\boldsymbol{q}_t^M = \boldsymbol{N}^M \boldsymbol{z}_t, \qquad \boldsymbol{N}^M \equiv \frac{1}{2}\left(\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma}\right)^{-1}. \qquad (38)
$$

From Equation (33), aggregate real profit is

$$
\Pi_t^M = \boldsymbol{q}_t^T \boldsymbol{\Sigma} \boldsymbol{q}_t = \boldsymbol{z}_t^T \boldsymbol{P}^M \boldsymbol{z}_t, \qquad \boldsymbol{P}^M \equiv (\boldsymbol{N}^M)^T \boldsymbol{\Sigma} \boldsymbol{N}^M.
$$

Output and Producer Surplus

As in the competitive equilibrium,

$$\boldsymbol{b}_t = \boldsymbol{z}_t - \frac{\zeta}{L}\boldsymbol{J}\boldsymbol{q}_t.$$

Using Equation (37), this becomes

$$\boldsymbol{b}_t = \left(\frac{\zeta}{L}\boldsymbol{J} + 2\boldsymbol{\Sigma}\right)\boldsymbol{q}_t.$$

Therefore, from Equation (1), monopoly output is

$$Y_t^M = \boldsymbol{q}_t^T\left(\frac{\zeta}{L}\boldsymbol{J} + 2\boldsymbol{\Sigma}\right)\boldsymbol{q}_t - \frac{1}{2}\boldsymbol{q}_t^T\boldsymbol{\Sigma}\boldsymbol{q}_t = \frac{1}{2}\boldsymbol{q}_t^T\left(2\frac{\zeta}{L}\boldsymbol{J} + 3\boldsymbol{\Sigma}\right)\boldsymbol{q}_t = \boldsymbol{z}_t^T\boldsymbol{Q}^M\boldsymbol{z}_t,$$

where

$$\boldsymbol{Q}^M \equiv \frac{1}{2}(\boldsymbol{N}^M)^T\left(2\frac{\zeta}{L}\boldsymbol{J} + 3\boldsymbol{\Sigma}\right)\boldsymbol{N}^M.$$

Dynamic R&D

The monopolist chooses R&D to maximize aggregate producer value:

$$V^{P,MM}(\boldsymbol{z}_0) \equiv \max_{\{\boldsymbol{x}_t\}_{t\geq 0}} \mathbb{E}_0\left[\int_0^\infty \exp\left(-\rho t\right)\left\{\boldsymbol{z}_t^T\boldsymbol{P}^M\boldsymbol{z}_t - \boldsymbol{x}_t^T\boldsymbol{x}_t\right\}dt\right],$$

subject to the knowledge accumulation law in Equation (4). We obtain the monopoly R&D rule by solving Equation (15) with $\frac{1}{2}\boldsymbol{N}^*$ replaced by $\boldsymbol{P}^M$. Let $\boldsymbol{X}^{P,MM}$ denote the symmetric coefficient matrix in the monopolist's producer-value function. The R&D allocation is

$$\boldsymbol{x}_t = \mu\boldsymbol{X}^{P,MM}\boldsymbol{z}_t,$$

and the induced transition is

$$\boldsymbol{\Phi}^{MM} \equiv \boldsymbol{\Omega} - \delta\boldsymbol{I} + \mu^2\boldsymbol{X}^{P,MM}.$$

Household Welfare

In contrast to the social planner, the monopolist maximizes aggregate producer value rather than household utility. Hence household welfare under the monopoly policy is computed separately:

$$V^{MM}\left(z_0\right)=\mathbb{E}_0\left[\int_0^{\infty}\exp\left(-\rho t\right)\left(z_t^T\boldsymbol{Q}^M z_t-\boldsymbol{x}_t^T\boldsymbol{x}_t\right)dt\bigg|\, z_0\right],$$

subject to $dz_t=\boldsymbol{\Phi}^{MM}z_t dt+\gamma\operatorname{diag}\left(z_t\right)d\boldsymbol{W}_t$. Guessing $V^{MM}(z)=z^T\boldsymbol{X}^{MM}z$ gives the Lyapunov equation

$$\begin{aligned}0=\boldsymbol{Q}^M-\mu^2\left(\boldsymbol{X}^{P,MM}\right)^T\boldsymbol{X}^{P,MM}+\boldsymbol{X}^{MM}\left(\boldsymbol{\Phi}^{MM}-\frac{\rho}{2}\boldsymbol{I}\right)\\+\left(\boldsymbol{\Phi}^{MM}-\frac{\rho}{2}\boldsymbol{I}\right)^T\boldsymbol{X}^{MM}+\gamma^2\mathcal{D}\left(\boldsymbol{X}^{MM}\right).\end{aligned}\tag{39}$$

Therefore, $\boldsymbol{X}^{MM}$ is obtained as the solution of Equation (39), and welfare in the monopoly problem is $V^{MM}(z_0)=z_0^T\boldsymbol{X}^{MM}z_0$.

B.4 Uniform R&D Subsidy Equations

This appendix records the equations used in the uniform R&D subsidy exercise. Let $s\in[0,1)$ denote a subsidy rate that reduces the private cost of R&D effort from $x_{i,t}^2$ to $(1-s)x_{i,t}^2$. Holding the competitive static product-market mapping fixed, firm i solves

$$V^i\left(z_0\right)=\max_{\{x_{i,t}\}_{t\geq 0}}\mathbb{E}_0\left[\int_0^{\infty}e^{-\rho t}\left\{z_t^T\boldsymbol{Q}_i z_t-(1-s)x_{i,t}^2\right\}dt\right],$$

taking other firms' R&D rules as given. With the quadratic guess $V^i(z)=z^T\boldsymbol{X}^i z$, the first-order condition is

$$x_i=\frac{\mu}{1-s}\left(\boldsymbol{X}_i^i\right)^T z.$$

Stacking firms' policies into the subsidy-specific policy matrix $\widetilde{\boldsymbol{X}}(s)$ gives

$$\boldsymbol{x}_t=\frac{\mu}{1-s}\widetilde{\boldsymbol{X}}(s)z_t,\qquad \boldsymbol{\Phi}(s)=\boldsymbol{\Omega}-\delta\boldsymbol{I}+\frac{\mu^2}{1-s}\widetilde{\boldsymbol{X}}(s).$$

The competitive Riccati equations under the subsidy become

$$0=\boldsymbol{Q}_i-\frac{\mu^2}{1-s}\boldsymbol{X}_i^i\left(\boldsymbol{X}_i^i\right)^T+\left(\boldsymbol{\Phi}(s)-\frac{\rho}{2}\boldsymbol{I}\right)^T\boldsymbol{X}^i+\boldsymbol{X}^i\left(\boldsymbol{\Phi}(s)-\frac{\rho}{2}\boldsymbol{I}\right)+\gamma^2\mathcal{D}\left(\boldsymbol{X}^i\right).$$

The subsidy changes firms' private cost but not the resource cost borne by the household.

Hence household welfare subtracts the full cost

$$\boldsymbol{x}_t^T \boldsymbol{x}_t = \frac{\mu^2}{(1-s)^2} \boldsymbol{z}_t^T \widetilde{\boldsymbol{X}}(s)^T \widetilde{\boldsymbol{X}}(s) \boldsymbol{z}_t.$$

Given the equilibrium policy, household welfare is obtained from the Lyapunov equation

$$\begin{aligned} 0 = \boldsymbol{Q} - \frac{\mu^2}{(1-s)^2} \widetilde{\boldsymbol{X}}(s)^T \widetilde{\boldsymbol{X}}(s) + \boldsymbol{X}(s) \left(\boldsymbol{\Phi}(s) - \frac{\rho}{2} \boldsymbol{I} \right) \\ + \left(\boldsymbol{\Phi}(s) - \frac{\rho}{2} \boldsymbol{I} \right)^T \boldsymbol{X}(s) + \gamma^2 \mathcal{D} \left(\boldsymbol{X}(s) \right). \end{aligned}$$

The numerical exercise solves this system for $s = 0, 0.01, \ldots, 0.50$ and selects the value of s that maximizes $\boldsymbol{z}_0^T \boldsymbol{X}(s) \boldsymbol{z}_0$ at the 2017 knowledge-capital vector.

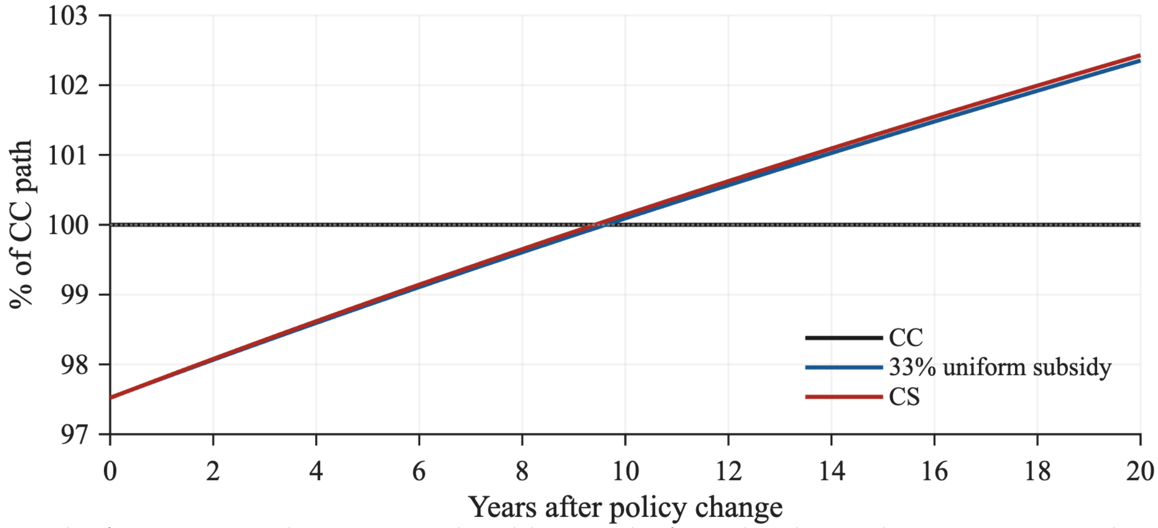
Table 8 reports the allocation comparison underlying the uniform-subsidy result. Since the subsidy changes a common proportional R&D price, the table sorts firms by the baseline local first-order-condition subsidy, $s_i^{\text{loc}}(\boldsymbol{z}) = 1 - \text{PMR}_i(\boldsymbol{z})/\text{SMR}_i(\boldsymbol{z})$, rather than by the additive level wedge used in Figure 4.

Table 8: R&D Resource-Cost Shares by Baseline Local-Subsidy Decile

Decile	Baseline local Median subsidy (%)	R&D resource-cost share (%) Competitive	Uniform subsidy	Planner (CS)
1	-71.8	0.5	0.4	0.1
2	-6.1	1.2	1.2	0.5
3	9.1	1.7	1.6	0.9
4	16.3	2.4	2.3	1.4
5	21.2	3.4	3.3	2.4
6	25.7	4.4	4.4	3.5
7	29.4	7.5	7.4	6.6
8	32.5	18.3	18.3	17.9
9	35.3	41.7	42.0	44.6
10	39.8	18.9	19.0	22.2

Notes: Deciles sort firms by the baseline local subsidy, $s_i^{\text{loc}} = 1 - \text{PMR}_i/\text{SMR}_i$, among the 739 firms with positive private and social marginal returns. Negative values indicate a local tax. R&D resource-cost shares are the squared-effort shares $x_i^2 / \sum_j x_j^2$ at the observed 2017 state. "Uniform" is the solved 33% competitive-subsidy equilibrium. "Planner" is the constrained-planner allocation (CS), evaluated under Cournot product-market conduct.

Figure 7: Short-Run Consumption-Flow Paths under R&D Subsidy and Constrained Planner



Notes: The figure reports deterministic closed-loop paths from the observed 2017 state, normalized by the contemporaneous competitive path so that CC equals 100 at each horizon. The subsidy path uses the grid-optimal 33% uniform R&D subsidy. Subsidy payments are lump-sum financed; the initial decline reflects higher R&D resource costs.

Because scale and allocation interact, the fraction of the CS welfare gain assigned to scale depends on the order of the two changes. Moving scale first assigns 90% of the gain to scale, while moving allocation first assigns 96%; the Shapley average assigns 93% to scale and 7% to allocation. At the firm level, let $r_i^k = (x_i^k)^2 / \sum_j (x_j^k)^2$ denote firm i's R&D resource-cost share under allocation k. The total-variation distance from the CS shares is 0.0629 under competition and 0.0582 under the 33% subsidy. The subsidy-induced and planner-induced changes in these shares have a correlation of 0.840. These diagnostics show that the subsidy primarily expands aggregate R&D and only modestly shifts its firm-level allocation.

B.5 Social-Private R&D Wedge Decomposition

This appendix gives the component definitions used in Figure 4. Let $\boldsymbol{e}_i$ denote the ith standard basis vector. Let $\boldsymbol{P}$ denote the aggregate competitive gross-profit matrix,

$$\boldsymbol{P} \equiv \sum_{j=1}^{n} \boldsymbol{Q}_j = \boldsymbol{N}^T \boldsymbol{N}.$$

Thus aggregate competitive gross profit is $\sum_j \pi_j(\boldsymbol{z}) = \boldsymbol{z}^T \boldsymbol{P} \boldsymbol{z}$. The social-private R&D wedge can be written as

$$\Delta_i(\boldsymbol{z}) = 2\mu\, \boldsymbol{e}_i^T \left(\boldsymbol{X} - \boldsymbol{X}^i \right) \boldsymbol{z}. \tag{40}$$

Define $\boldsymbol{M}_i \equiv \boldsymbol{X} - \boldsymbol{X}^i$ and

$$\boldsymbol{C}_{-i} \equiv \widetilde{\boldsymbol{X}}^T \left(\boldsymbol{I} - \boldsymbol{e}_i \boldsymbol{e}_i^T \right) \widetilde{\boldsymbol{X}}.$$

Since the competitive feedback rule is $\boldsymbol{x}^C(\boldsymbol{z}) = \mu \widetilde{\boldsymbol{X}} \boldsymbol{z}$, the quadratic form $\mu^2 \boldsymbol{C}_{-i}$ gives the R&D resource costs of all firms other than i under the competitive policy. Subtracting firm i's Riccati equation from the household welfare Lyapunov equation gives

$$\boldsymbol{0} = \boldsymbol{F}_i + \boldsymbol{M}_i \left(\boldsymbol{\Phi} - \frac{\rho}{2} \boldsymbol{I} \right) + \left(\boldsymbol{\Phi} - \frac{\rho}{2} \boldsymbol{I} \right)^T \boldsymbol{M}_i + \gamma^2 \mathcal{D}(\boldsymbol{M}_i), \tag{41}$$

where the flow wedge is

$$\begin{aligned} \boldsymbol{F}_i &\equiv \boldsymbol{Q} - \boldsymbol{Q}_i - \mu^2 \boldsymbol{C}_{-i} \\ &= \underbrace{\frac{1}{2} \boldsymbol{N}^T \boldsymbol{\Sigma} \boldsymbol{N}}_{\boldsymbol{F}^{\mathrm{CS}}} + \underbrace{\frac{\zeta}{L} \boldsymbol{N}^T \boldsymbol{J} \boldsymbol{N}}_{\boldsymbol{F}^{\mathrm{LI}}} + \underbrace{\boldsymbol{P} - \boldsymbol{Q}_i}_{\boldsymbol{F}_i^{\mathrm{RP}}} + \underbrace{\left(-\mu^2 \boldsymbol{C}_{-i} \right)}_{\boldsymbol{F}_i^{\mathrm{RRC}}}. \end{aligned} \tag{42}$$

This expression directly separates the flow-payoff consequences of a marginal increase in firm i's knowledge drift under the competitive closed-loop policy. The first term is aggregate consumer surplus under linear demand, $\frac{1}{2} \boldsymbol{q}^T \boldsymbol{\Sigma} \boldsymbol{q}$, expressed as a quadratic form in knowledge capital. The second is aggregate labor income, $(\zeta/L) \boldsymbol{q}^T \boldsymbol{J} \boldsymbol{q}$. Together they satisfy

$$\boldsymbol{F}^{\mathrm{CS}} + \boldsymbol{F}^{\mathrm{LI}} = \boldsymbol{Q} - \boldsymbol{P}$$

and constitute the appropriability gap: household flow value generated by innovation but not captured in aggregate producer gross profit.

Since $\boldsymbol{P} = \boldsymbol{N}^T \boldsymbol{N}$, consumer surplus satisfies

$$\boldsymbol{F}^{\mathrm{CS}} = \frac{1}{2} \boldsymbol{P} + \frac{1}{2} \boldsymbol{N}^T (\boldsymbol{\Sigma} - \boldsymbol{I}) \boldsymbol{N}. \tag{43}$$

Thus, linear demand does not generally make consumer surplus proportional to aggregate producer gross profit. In the no-cross-product benchmark $\boldsymbol{\Sigma} = \boldsymbol{I}$, the second term is zero; in the estimated network economy, it need not vanish.

The rival-profit flow $\boldsymbol{F}_i^{\mathrm{RP}} = \boldsymbol{P} - \boldsymbol{Q}_i$ is the gross-profit flow accruing to firms other than i; the reported RP component is its marginal value with respect to z_i, which is negative when higher z_i lowers rivals' quantities. The flow $\boldsymbol{F}_i^{\mathrm{RRC}} = -\mu^2 \boldsymbol{C}_{-i}$ subtracts the R&D resource costs borne by other firms under their competitive feedback rules. The sign of the resulting RRC marginal component depends on the closed-loop feedback response; the positive average entries in Figure 4 indicate that the marginal state change lowers discounted rival R&D resource costs in the baseline decile groups.

For each component $m \in \{\mathrm{CS}, \mathrm{LI}, \mathrm{RP}, \mathrm{RRC}\}$, solve Equation (41) with the corresponding flow matrix. Denote the solution by $\boldsymbol{M}_i^m$, suppressing the firm subscript when the flow matrix is common across firms. The marginal-return component is

$$\Delta_i^m(\boldsymbol{z}) = 2\mu\, \boldsymbol{e}_i^T \boldsymbol{M}_i^m \boldsymbol{z}. \tag{44}$$

Accordingly, firm i's CS component is the marginal effect of its R&D on discounted aggregate consumer surplus, not a static allocation of contemporaneous CS across products. This derivative-based definition matches the object entering the firm's social marginal R&D return. Because Equation (41) is linear in $(\boldsymbol{M}, \boldsymbol{F})$, the flow decomposition in Equation (42) carries through to discounted marginal values:

$$\Delta_i = \Delta_i^{\mathrm{CS}} + \Delta_i^{\mathrm{LI}} + \Delta_i^{\mathrm{RP}} + \Delta_i^{\mathrm{RRC}}. \tag{45}$$

These terms provide a payoff-flow accounting. Technology spillovers operate through the closed-loop state dynamics and the solved R&D feedback rule, thereby shaping every discounted component.

B.6 Instantaneous Expected Log Output Growth

This subsection derives the instantaneous expected log-output growth formula used in the baseline decomposition below. Apply Itô's lemma to the closed-loop knowledge law

$$d\boldsymbol{z}_t = \boldsymbol{\Phi} \boldsymbol{z}_t dt + \gamma \operatorname{diag}(\boldsymbol{z}_t)\, d\boldsymbol{W}_t$$

and to output $Y_t = f(\boldsymbol{z}_t) = \boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t$. The matrix $\boldsymbol{Q}$ is symmetric by construction, and the log-growth formula below is evaluated on states with $Y_t > 0$. The required derivatives are

$$\begin{aligned} f_z(\boldsymbol{z}_t) &= 2\boldsymbol{Q}\boldsymbol{z}_t, \\ f_{zz}(\boldsymbol{z}_t) &= 2\boldsymbol{Q}. \end{aligned}$$

Therefore,

$$
\begin{aligned}
dY_t &= \left\{2\boldsymbol{z}_t^T \boldsymbol{Q}\boldsymbol{\Phi}\boldsymbol{z}_t + \frac{1}{2}\mathrm{Tr}\left[2\gamma^2 \operatorname{diag}(\boldsymbol{z}_t)\,\boldsymbol{Q}\operatorname{diag}(\boldsymbol{z}_t)\right]\right\} dt + 2\gamma \boldsymbol{z}_t^T \boldsymbol{Q}\operatorname{diag}(\boldsymbol{z}_t)\, d\boldsymbol{W}_t \\
&= \left\{\boldsymbol{z}_t^T \left(\boldsymbol{Q}\boldsymbol{\Phi} + \boldsymbol{\Phi}^T\boldsymbol{Q}\right)\boldsymbol{z}_t + \gamma^2 \sum_i z_{i,t}^2 \boldsymbol{Q}_{ii}\right\} dt + 2\gamma \boldsymbol{z}_t^T \boldsymbol{Q}\operatorname{diag}(\boldsymbol{z}_t)\, d\boldsymbol{W}_t
\end{aligned}
$$

Next, applying Itô's lemma to $g(Y_t) = \log Y_t$ and using the diffusion term in dY_t gives

$$
\begin{aligned}
d\log Y_t =& \left[\frac{\boldsymbol{z}_t^T\left(\boldsymbol{Q}\boldsymbol{\Phi} + \boldsymbol{\Phi}^T\boldsymbol{Q}\right)\boldsymbol{z}_t}{\boldsymbol{z}_t^T\boldsymbol{Q}\boldsymbol{z}_t} + \gamma^2\left\{\frac{\sum_i z_{i,t}^2 \boldsymbol{Q}_{ii}}{\boldsymbol{z}_t^T\boldsymbol{Q}\boldsymbol{z}_t} - 2\frac{\boldsymbol{z}_t^T\boldsymbol{Q}\operatorname{diag}\left(\boldsymbol{z}_t^2\right)\boldsymbol{Q}\boldsymbol{z}_t}{\left(\boldsymbol{z}_t^T\boldsymbol{Q}\boldsymbol{z}_t\right)^2}\right\}\right] dt \\
&+ \frac{2\gamma}{\boldsymbol{z}_t^T\boldsymbol{Q}\boldsymbol{z}_t}\boldsymbol{z}_t^T\boldsymbol{Q}\operatorname{diag}(\boldsymbol{z}_t)\, d\boldsymbol{W}_t.
\end{aligned}
$$

Because the technology transition matrix can be written as

$$
\boldsymbol{\Phi} = \boldsymbol{\Omega} - \delta\boldsymbol{I} + \mu^2\widetilde{\boldsymbol{X}},
$$

the first drift term in log output can be decomposed as

$$
\boldsymbol{Q}\boldsymbol{\Phi} + \boldsymbol{\Phi}^T\boldsymbol{Q} = \left(\boldsymbol{Q}\boldsymbol{\Omega} + \boldsymbol{\Omega}^T\boldsymbol{Q}\right) + \mu^2\left(\boldsymbol{Q}\widetilde{\boldsymbol{X}} + \widetilde{\boldsymbol{X}}^T\boldsymbol{Q}\right) - 2\delta\boldsymbol{Q}.
$$

Thus, the contributions of technology spillovers, R&D, obsolescence, and the Itô term to instantaneous 2017 g_Y are

$$
\frac{d\log Y_t|_{\text{spillover}}}{dt} = \frac{\boldsymbol{z}_t^T\left(\boldsymbol{Q}\boldsymbol{\Omega} + \boldsymbol{\Omega}^T\boldsymbol{Q}\right)\boldsymbol{z}_t}{\boldsymbol{z}_t^T\boldsymbol{Q}\boldsymbol{z}_t} \tag{46}
$$

$$
\frac{d\log Y_t|_{\text{R\&D}}}{dt} = \mu^2\frac{\boldsymbol{z}_t^T\left(\boldsymbol{Q}\widetilde{\boldsymbol{X}} + \widetilde{\boldsymbol{X}}^T\boldsymbol{Q}\right)\boldsymbol{z}_t}{\boldsymbol{z}_t^T\boldsymbol{Q}\boldsymbol{z}_t} \tag{47}
$$

$$
\frac{d\log Y_t|_{\text{Obsolescence}}}{dt} = -2\delta \tag{48}
$$

$$
\frac{d\log Y_t|_{\text{Itô}}}{dt} = \gamma^2\left\{\frac{\sum_i z_{i,t}^2 \boldsymbol{Q}_{ii}}{\boldsymbol{z}_t^T\boldsymbol{Q}\boldsymbol{z}_t} - 2\frac{\boldsymbol{z}_t^T\boldsymbol{Q}\operatorname{diag}\left(\boldsymbol{z}_t^2\right)\boldsymbol{Q}\boldsymbol{z}_t}{\left(\boldsymbol{z}_t^T\boldsymbol{Q}\boldsymbol{z}_t\right)^2}\right\}.
$$

B.7 Baseline Instantaneous Growth Decomposition

To interpret the g_Y column in the counterfactual table, we decompose the baseline instantaneous 2017 output-growth rate into technology spillovers, obsolescence, R&D, and the contribution of shocks. The components are computed from the primitive matrices $\boldsymbol{Q}$, $\boldsymbol{\Omega}$, $\widetilde{\boldsymbol{X}}$, and the observed 2017 knowledge-capital vector so that they add up to g_Y under the baseline policy.

The baseline instantaneous 2017 output-growth rate of 1.50% is the net of a 3.21 percentage-point contribution from technology spillovers, a 0.52 point contribution from direct R&D, and a 2.23 point drag from knowledge obsolescence. In the baseline deterministic specification, the Itô term is zero because $\gamma = 0$. Changing R&D incentives affects g_Y through the endogenous R&D component, while changing product-market allocations changes the output matrix $\boldsymbol{Q}$ and hence how knowledge capital maps into aggregate output.

B.8 Alternative Technology-Proximity Measure

The Jaffe measure treats patent classes as orthogonal dimensions. As a measurement check, we instead use a Mahalanobis measure that allows classes to be related when they tend to co-occur across firms. We construct the class-co-location matrix from the same firm patent-share vectors, use it to weight pairwise overlap, set the diagonal to zero, and normalize each recipient row.

Table 9 compares the resulting estimates with the Jaffe estimates. The Mahalanobis technology-exposure coefficient is positive and statistically significant in all three specifications: the baseline, the product-market-exposure horse race, and the specification with technology-section-by-year fixed effects.

B.9 Spillover-Intensity Horizon Robustness

Table 10 uses cumulative three-year knowledge-capital growth as the dependent variable in the same three OLS specifications. All three estimates are positive; those in specifications (2) and (3) are statistically significant at the 10% level. These cumulative coefficients provide a persistence check for the one-year panel relationship.

B.10 Spillover-Intensity Sensitivity

Table 11 reports results from solving the model separately at $\beta \in \{0.01, 0.02, 0.03\}$, with μ and δ recalibrated and the R&D policies and counterfactuals recomputed at each value.

Table 9: Technology-Proximity Measure Robustness

	(1)	(2)	(3)
Panel A: Main-group Jaffe			
Technology exposure	0.039***	0.021**	0.025***
	(0.010)	(0.008)	(0.008)
Product-market exposure		-0.008***	-0.009***
		(0.002)	(0.002)
Panel B: Main-group Mahalanobis			
Technology exposure	0.091***	0.051***	0.065***
	(0.019)	(0.015)	(0.015)
Product-market exposure		-0.007***	-0.008***
		(0.002)	(0.002)
Controls: x_i/z_i, $\log z_i$	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes
Product-market exposure	No	Yes	Yes
Year/interacted fixed effects	Year	Year	Tech-year
Observations	18,724	18,724	18,724

Notes: The dependent variable is $\Delta \log z_i$. Both panels use CPC main groups and the centered five-year patent window. Panel A uses Jaffe overlap. Panel B weights overlap by the Mahalanobis class-co-location matrix before setting the diagonal to zero and normalizing each recipient row. Columns reproduce the three specifications in Table 3. Standard errors, two-way clustered by firm and calendar year, are in parentheses. Variables use the same 0.5 percent winsorization rule as in the main table. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 10: Horizon Robustness of Spillover-Intensity Estimates

	Technology-exposure coefficient	
	$h = 1$ (annual)	$h = 3$ (cumulative)
(1) Ordinary least squares + controls	0.039***	0.070***
	(0.010)	(0.023)
(2) + Product-market exposure	0.021**	0.037*
	(0.008)	(0.021)
(3) + Tech-section × year fixed effects	0.025***	0.037*
	(0.008)	(0.020)
Observations	18,724	14,847

Notes: The dependent variable is cumulative knowledge-capital growth, $\log z_{i,t+h} - \log z_{i,t}$. Rows reproduce columns (1)–(3) of Table 3. The $h = 3$ estimates report cumulative three-year growth. Standard errors, two-way clustered by firm and calendar year, are in parentheses. Variables use the same 0.5 percent winsorization rule as in the main table. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 11: Spillover-Intensity Sensitivity

β	μ	δ	CC g_Y components, pp			Welfare		Uniform subsidy	
			Spill.	R&D	Dep.	CS welfare	SS welfare	Subsidy rate (%)	Subsidy welfare
0.010	0.0473	0.0029	1.61	0.47	-0.58	100.37	109.99	29	100.32
0.020	0.0524	0.0112	3.21	0.52	-2.23	100.50	108.97	33	100.45
0.030	0.0572	0.0194	4.82	0.56	-3.88	100.76	108.38	37	100.68

Notes: For each reported β, μ and δ are recalibrated in the 2017 competitive equilibrium to match aggregate R&D intensity of 2.6% and instantaneous 2017 output growth g_Y of 1.5%. Spill., R&D, and Dep. decompose competitive-equilibrium g_Y in percentage points. CS, SS, and uniform-subsidy welfare are normalized to the competitive equilibrium at the same β. The uniform-subsidy rate is the grid-optimal R&D subsidy on the 0–50% grid.

Table 12: Product-Market Parameter Sensitivity

		Scenario outcomes (CC=100)			Uniform subsidy	
α	Median SMR/PMR	CS welfare	SS output	SS welfare	Rate (%)	Welfare (CC=100)
0.08	1.319	100.53	109.86	109.50	33	100.49
0.12	1.305	100.50	109.29	108.97	33	100.45
0.16	1.296	100.47	108.79	108.51	32	100.42

Notes: For each α, the 2017 knowledge-capital state is recovered from accounting profits using the corresponding product-market matrix, and μ and δ are recalibrated in the competitive equilibrium to match aggregate R&D intensity of 2.6% and instantaneous 2017 g_Y of 1.5%. The spillover parameter is fixed at $\beta = 0.02$ and the discount rate at $\rho = 0.10$. Median SMR/PMR is the median of $\mathrm{SMR}_i(z)/\mathrm{PMR}_i(z)$ among firms with positive private and social marginal R&D returns. CS welfare, SS output, SS welfare, and subsidy welfare are normalized to no-subsidy competition at the same α. The subsidy rate is the welfare-maximizing uniform R&D subsidy on the 0–50% grid with 1 percentage point spacing.

B.11 Product-Market Parameter Sensitivity

Table 12 varies the product-market parameter around the baseline value while keeping the central spillover parameter fixed at $\beta = 0.02$. For each value, the 2017 knowledge-capital state is recovered with the corresponding product-market matrix and the dynamic parameters are recalibrated to the same R&D-intensity and instantaneous 2017 output-growth targets. As α rises from 0.08 to 0.16, the CS welfare gain declines from 0.53% to 0.47%, the SS welfare gain declines from 9.50% to 8.51%, and the grid-optimal uniform subsidy changes from 33% to 32%.

B.12 Discount-Rate Sensitivity

Table 13 varies the discount rate and recalibrates the dynamic parameters to the same R&D-intensity and instantaneous 2017 output-growth targets. Lower discount rates put more

Table 13: Discount-Rate Sensitivity with Recalibrated Dynamic Parameters

ρ	μ	δ	Min. stability $\rho - 2g_{\max}$	CM	CS	MM	SS	Opt. sub. (%)	Opt. sub. welfare
					Welfare (CC=100)				
5.0	0.0270	0.0099	0.0033	98.26	101.81	95.66	108.41	46	101.49
7.5	0.0398	0.0105	0.0335	98.71	100.71	95.60	108.45	36	100.64
10.0	0.0524	0.0112	0.0559	98.91	100.50	95.66	108.97	33	100.45

Notes: For each discount rate, μ and δ are recalibrated in the competitive 2017 equilibrium to match aggregate R&D intensity of 2.6% and instantaneous 2017 g_Y of 1.5%. Welfare columns are normalized to the competitive equilibrium at the same ρ. Min. stability is the minimum of $\rho - 2g_{\max}$ across the solved main scenarios, where $g_{\max}$ is the largest real eigenvalue of the closed-loop transition matrix $\mathbf{\Phi}$; positive values indicate finite discounted values. Opt. sub. is the welfare-maximizing uniform R&D subsidy on the 0–50% grid, and opt. sub. welfare is normalized to no-subsidy competition at the same ρ.

weight on future growth and therefore raise the welfare value of R&D interventions. The main scenario ranking is preserved across $\rho \in \{5\%, 7.5\%, 10\%\}$. The minimum finite-value margin remains positive throughout but falls to 0.0033 at $\rho = 5\%$, so that case lies close to the stability boundary.

Appendix C Additional Quantitative Results

C.1 R&D Effort, Social-to-Private Marginal Returns, and Network Centrality

Table 14 reports 2017 cross-sectional regressions of the logs of model-implied private R&D effort, constrained-planner effort, and observed R&D effort, and of the log social-to-private marginal-return ratio, on log firm knowledge and the two network-centrality measures. All columns use the same 710 firms. Product-market and technology-spillover centrality are the max-normalized left-eigenvector centralities of $\mathbf{\Sigma}$ and $\mathbf{\Omega}$, respectively, after setting their diagonals to zero. Conditional on log firm knowledge and the other centrality measure, the return ratio rises with technology-spillover centrality and falls with product-market centrality.

The positive coefficient on $\log z_i$ in the return-ratio regression is consistent with the model's static markup mapping. In the Cournot allocation, markup is increasing in the firm's equilibrium quantity share, and quantities satisfy $\boldsymbol{q} = \boldsymbol{N}\boldsymbol{z}$. In the estimated 2017 economy, the correlation between $\log z_i$ and the Lerner index is 0.889.

Because $\log(\mathrm{SMR}_i/\mathrm{PMR}_i) = \log \mathrm{SMR}_i - \log \mathrm{PMR}_i$, each return-ratio coefficient is the difference between the corresponding social- and private-return coefficients. The private R&D first-order condition implies $\mathrm{PMR}_i = 2x_i$, so the first-column coefficient on $\log z_i$,

Table 14: R&D Effort, Social-to-Private Marginal-Return Ratio, and Network Centrality

	Competitive log effort	Constrained planner log effort	Observed log effort	log(SMR/PMR)
$\log z$	3.32***	3.86***	3.60***	0.460***
	(0.089)	(0.154)	(0.135)	(0.063)
Product-market centrality	−1.73***	−2.24***	−0.066	−0.467***
	(0.061)	(0.129)	(0.088)	(0.062)
Technology-spillover centrality	0.243***	1.14***	0.462***	0.823***
	(0.068)	(0.124)	(0.124)	(0.061)
Observations	710	710	710	710
R^2	0.875	0.704	0.719	0.337

Notes: The table reports 2017 cross-sectional OLS regressions on a common sample of 710 firms with positive model-implied private effort, constrained-planner effort, observed R&D, and private and social marginal returns. Observed effort is $\sqrt{\text{Compustat R\&D}}$ because model R&D resource cost is x_i^2. Product-market and technology-spillover centralities are max-normalized left eigenvectors of the zero-diagonal $\mathbf{\Sigma}$ and $\mathbf{\Omega}$. All regressions include an intercept. Heteroskedasticity-consistent HC1 standard errors are in parentheses. *, **, and *** denote significance at 10%, 5%, and 1%.

3.322, is also the coefficient in the $\log \mathrm{PMR}_i$ regression. Using the same 710 firms and regressors, the corresponding coefficient in the $\log \mathrm{SMR}_i$ regression is 3.782, giving $3.782 - 3.322 = 0.460$. Thus both marginal returns rise strongly with knowledge, but the social return rises slightly faster.

Table 15 identifies the payoff-accounting sources of these return-ratio coefficients. For each wedge component Δ_i^m, $m \in \{\mathrm{CS}, \mathrm{LI}, \mathrm{RP}, \mathrm{RRC}\}$, define its exact log contribution as

$$\ell_i^m \equiv \frac{\Delta_i^m}{\mathrm{PMR}_i} \frac{\log(1+r_i)}{r_i}, \qquad r_i \equiv \frac{\Delta_i}{\mathrm{PMR}_i}.$$

Because the four components sum to Δ_i, their log contributions sum firm by firm to $\log(\mathrm{SMR}_i/\mathrm{PMR}_i)$. Their regression coefficients therefore add exactly to the total-wedge coefficient. These outcomes measure each component's signed contribution to the social return relative to the private marginal return.

For $\log z_i$, the total coefficient of 0.460 decomposes into 1.027 from CS, −3.396 from LI, 3.000 from RP, and −0.171 from RRC. LI rises in levels with firm knowledge, while its normalized contribution to the log wedge declines as private marginal returns expand. Because RP is negative on average, its positive coefficient indicates that its downward contribution becomes less important relative to private marginal returns as knowledge rises. The positive CS contribution and the attenuation of the negative RP contribution together outweigh the declining normalized LI and RRC contributions, producing the positive coefficient on the total wedge.

Table 15: Centrality and the Exact Log-Wedge Contributions

	CS	LI	RP	RRC	Total wedge
$\log z$	1.027***	−3.396***	3.000***	−0.171***	0.460***
	(0.095)	(0.237)	(0.246)	(0.014)	(0.063)
Product-market centrality	1.144***	0.682***	−2.430***	0.138***	−0.467***
	(0.069)	(0.177)	(0.244)	(0.014)	(0.062)
Technology-spillover centrality	−0.244***	−0.017	1.162***	−0.079***	0.823***
	(0.065)	(0.155)	(0.206)	(0.012)	(0.061)
Observations	710	710	710	710	710
R^2	0.500	0.361	0.342	0.342	0.337

Notes: The table reports 2017 cross-sectional OLS regressions on a common sample of 710 firms with positive private and social marginal returns and positive private-model, social-planner, and observed R&D effort. All three predictors enter jointly, and all regressions include an intercept. Product-market and technology-spillover centralities are max-normalized left eigenvectors of $\mathbf{\Sigma}$ and $\mathbf{\Omega}$, respectively. For wedge component Δ_i^m and $r_i = \Delta_i/\mathrm{PMR}_i$, its exact log contribution is $\ell_i^m = (\Delta_i^m/\mathrm{PMR}_i)\log(1+r_i)/r_i$. The component outcomes measure signed contributions to the social return relative to the private marginal return, and their coefficients add exactly to the total-wedge coefficient. Heteroskedasticity-consistent HC1 standard errors are in parentheses. *, **, and *** denote significance at 10%, 5%, and 1%.

The two centrality coefficients display a different pattern. The product-market-centrality coefficient of −0.467 decomposes into 1.144 from CS, 0.682 from LI, −2.430 from RP, and 0.138 from RRC. Thus the more negative rival-profit contribution more than offsets the positive CS and LI contributions. By contrast, the technology-centrality coefficient of 0.823 combines −0.244 from CS, −0.017 from LI, 1.162 from RP, and −0.079 from RRC. Because RP is negative on average, its positive coefficient captures a smaller rival-profit drag at higher technology centrality. The opposite associations of the two centrality measures with the total wedge therefore arise primarily from how the relative rival-profit contribution varies across firms. This rival-profit margin is the principal accounting force linking network position to the social-private return ratio in these regressions.

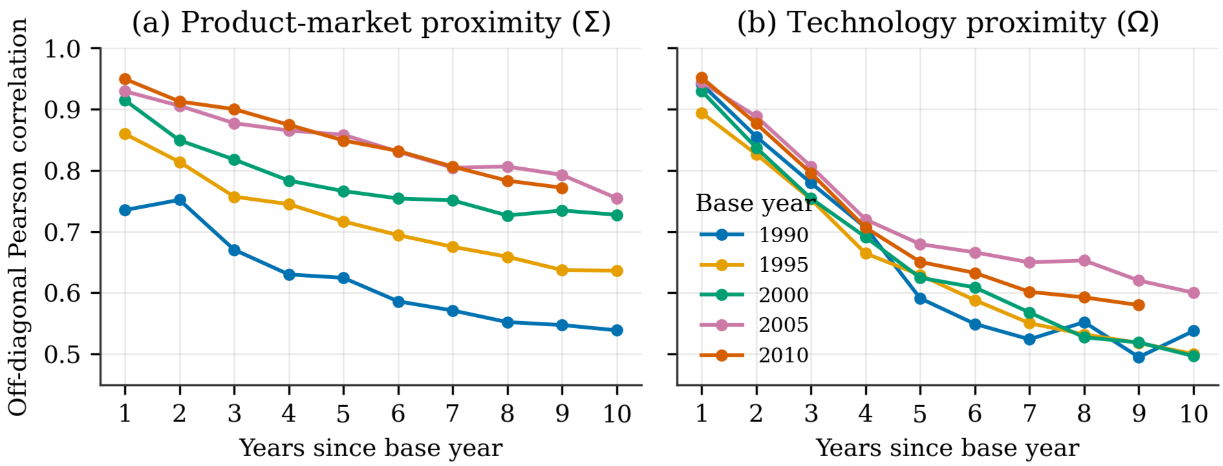
C.2 Network Persistence

Figure 8 documents the persistence of the two network primitives. For each base year, we compare off-diagonal entries with the corresponding entries h years later among firms observed in both years. Both matrices are highly persistent at short horizons, but the correlations decline with horizon length.

C.3 Computational Scale

The competitive problem requires n coupled $n \times n$ value-function matrices, whereas the monopoly and planner problems each require a single Riccati equation. Household welfare

Figure 8: Persistence of Product-Market and Technology Networks



Notes: Each line fixes a base year and plots the Pearson correlation between the off-diagonal entries of the base-year matrix and the matrix h years later. Each comparison restricts the sample to firms observed in both years. The left panel reports the product-rivalry matrix; the right panel reports the technology-spillover exposure matrix.

Table 16: State Representation and Computational Scale

Model	State representation	Per-update arithmetic	Number of firms
Cavenaile et al. (2026)	Joint grid with k^n points	$O(k^n)$ per full-grid update	Up to 4
Our model	Continuous state; $O(n^3)$ stored coefficients	$O(n^4)$ per dense update	418–802

Notes: The state-representation column reports the size of the value representation. For the joint-grid model, $O(k^n)$ records the scaling required to update all grid points, rather than an implementation-specific operation count. Storage and operation counts for our model refer to the dense implementation used in the quantitative analysis.

and producer value are then obtained from Lyapunov equations.

The competitive dynamic problem stores n^3 coefficients in total. In the 2017 baseline, $n = 757$, or about 434 million coefficients. Dense products such as $\boldsymbol{\Phi}^T \boldsymbol{X}^i$ cost $O(n^3)$ for each firm, so a complete update costs $O(n^4)$.

Within existing endogenous growth frameworks, solving a dynamic oligopoly problem involving all publicly traded U.S. firms with patents is computationally challenging. A joint grid with k points for each of n firm states contains k^n points. For instance, Cavenaile et al. (2026) analyze oligopolies with up to four firms using a discrete-state approach. Our continuous-state formulation avoids a joint state grid; it stores $O(n^3)$ value-function coefficients and requires $O(n^4)$ arithmetic per dense Riccati update. Table 16 separates these features of state representation from per-update arithmetic.

Table 17: Riccati Equation and Stability Diagnostics

Scenario	Riccati residual (relative max norm)	$\max \mathrm{Re}(\mathbf{\Phi} - \rho \boldsymbol{I}/2)$	$\rho - 2g_{\max}$
CC	1.7×10^{-6}	-0.0375	0.0751
CM	1.8×10^{-6}	-0.0369	0.0738
CS	1.6×10^{-6}	-0.0319	0.0638
MM	1.4×10^{-6}	-0.0368	0.0735
SS	1.8×10^{-6}	-0.0279	0.0559
CC+s ($s = 0.33$)	1.6×10^{-6}	-0.0342	0.0684

Notes: The Riccati residual is the maximum absolute equation residual divided by the maximum of one and the absolute entries of the constituent equation terms. $g_{\max}$ is the largest real eigenvalue of $\mathbf{\Phi}$. Negative values of $\max \mathrm{Re}(\mathbf{\Phi} - \rho \boldsymbol{I}/2)$, equivalently positive $\rho - 2g_{\max}$, indicate the deterministic stabilizing condition used in Assumption 1. All reported rows satisfy this condition.

C.4 Numerical Solution and Equilibrium Selection

For the competitive dynamic problem, the numerical algorithm used in the reported quantitative exercises iterates on the coefficient matrices of firms' quadratic value functions in the deterministic specification, $\gamma = 0$. Let k denote the iteration index. Given the current firm-level value matrices $\boldsymbol{X}_k^1, \ldots, \boldsymbol{X}_k^n$, we recover $\widetilde{\boldsymbol{X}}_k$, construct the implied closed-loop matrix $\mathbf{\Phi}_k$, and update the matrices using the forward-Euler step

$$\frac{\boldsymbol{X}_{k+1}^i - \boldsymbol{X}_k^i}{\Delta} = \boldsymbol{Q}_i - \mu^2 \boldsymbol{X}_{i,k}^i \left(\boldsymbol{X}_{i,k}^i\right)^T + \left(\mathbf{\Phi}_k - \frac{\rho}{2}\boldsymbol{I}\right)^T \boldsymbol{X}_k^i + \boldsymbol{X}_k^i \left(\mathbf{\Phi}_k - \frac{\rho}{2}\boldsymbol{I}\right).$$

Here Δ is the pseudo-time step, and $\boldsymbol{X}_{i,k}^i$ is the ith column of firm i's value-function matrix. We initialize the no-subsidy system at $\boldsymbol{X}_0^i = \mathbf{0}$, which selects the stabilizing finite-horizon limit with zero terminal values when that limit exists (Cartigny and Michel, 2003). Subsidy counterfactuals use continuation across the subsidy grid. Iterations stop when the maximum absolute change in $\widetilde{\boldsymbol{X}}_k$ falls below the convergence tolerance. Table 17 reports the relative max norm of each deterministic algebraic Riccati equation and the associated stability margins. Across the reported allocations, these residuals do not exceed 1.8×10^{-6}, and the largest real eigenvalue of $\mathbf{\Phi} - \rho \boldsymbol{I}/2$ is at most -0.0279, so every reported solution satisfies the deterministic stability condition.

C.5 Feasibility of the Linear-Quadratic Allocation

This subsection examines feasibility at the observed 2017 state, along transition paths, and in the long-run balanced-growth limit. In the R&D truncation exercises, we hold the coefficients of the solved linear feedback rules fixed and replace negative prescribed effort

Table 18: Negative Quantities and Nonnegativity-Constrained Output at the Observed 2017 State

Scenario	Neg. firms	Neg. q^2 (%)	Closed-form output	$q \geq 0$ QP output	Output change	Zero-quantity firms
MM	69/757	0.07	96.33	96.27	-0.06	77/757
SS	234/757	0.79	109.29	108.82	-0.47	283/757

Notes: The table evaluates the closed-form static mapping $\boldsymbol{q}^u(\boldsymbol{z}) = \boldsymbol{N}\boldsymbol{z}$ at the observed 2017 knowledge-capital vector. Negative quantities arise only in the full static reallocation benchmarks MM and SS; the competitive static mapping (CC, CM, and CS) yields positive quantities. Negative q^2 is $\sum_{i:q_i^u<0}(q_i^u)^2/\sum_i(q_i^u)^2$. Firms with negative quantities account for at most 1.03 percent of observed sales and 2.58 percent of observed R&D. The quadratic-program (QP) columns solve the corresponding static nonnegative-quantity problem at the same state while holding the dynamic R&D policy fixed; output is normalized to CC=100.

with zero. The main scenario table reports the untruncated linear-quadratic policies.

Feasibility at the Observed State

Static quantities. Table 18 reports negative static quantities at the observed 2017 knowledge-capital vector together with the corresponding static nonnegative-quantity quadratic programs (QPs). The competitive static mapping (CC, CM, and CS) yields positive quantities. In the full static reallocation benchmarks, 69 firms in MM and 234 firms in SS have negative quantities, accounting for 0.07 and 0.79 percent of the quadratic quantity aggregate. At the same state and dynamic policy, the static QPs move output from 96.33 to 96.27 in MM and from 109.29 to 108.82 in SS.

R&D effort. Table 19 reports the number and severity of negative R&D effort at the observed 2017 knowledge-capital vector, together with the growth effect of truncating negative effort at zero. The competitive benchmark has positive R&D effort for every firm. Negative effort appears in some counterfactuals, especially CM, where 421 firms have negative effort and account for 5.11 percent of R&D resource cost; the corresponding shares are much smaller in the other counterfactuals. Without re-solving the dynamic game, we replace $\boldsymbol{x}^u(\boldsymbol{z})$ with $\boldsymbol{x}^+(\boldsymbol{z}) = \max\{\boldsymbol{0}, \boldsymbol{x}^u(\boldsymbol{z})\}$ in both the R&D resource-cost calculation and the knowledge drift. This changes instantaneous 2017 g_Y by less than 0.003 percentage points in every scenario.

Table 19: Negative R&D Effort and Growth after Truncation at the Observed 2017 State

Scenario	Neg. firms	R&D-cost share from $x < 0$ (%)	Sales (%)	Obs. R&D (%)	Effect of truncation on g_Y (pp)
CC	0/757	0.00	0.00	0.00	0.0000
CM	421/757	5.11	5.27	6.93	0.0014
CS	23/757	0.01	0.01	0.07	-0.0001
MM	228/757	0.95	1.36	1.68	-0.0007
SS	111/757	0.17	0.27	0.72	-0.0029

Notes: R&D-cost share from $x < 0$ is $\sum_{i:x_i<0} x_i^2 / \sum_i x_i^2$ under the linear feedback policy at the observed 2017 state. Sales and observed R&D are shares of observed 2017 totals accounted for by firms with negative R&D effort. The final column reports the change in instantaneous 2017 g_Y when $\boldsymbol{x}^+(\boldsymbol{z}) = \max\{\boldsymbol{0}, \boldsymbol{x}^u(\boldsymbol{z})\}$ is used in both R&D resource cost and the knowledge drift, holding the feedback coefficients fixed.

Transition-Path Feasibility

Table 20 extends the sign check along deterministic closed-loop paths from the observed 2017 state and reports the welfare effect of truncating negative R&D effort at zero along each path. Knowledge capital remains positive in all scenarios over the 50-year horizon. In the competitive benchmark, the first negative R&D effort appears after 24.5 years and the first negative static quantity after 33.3 years. Across the counterfactuals, the largest share of R&D resource cost attributable to firms with negative effort is 6.62% in CM, and the largest share of total q^2 attributable to firms with negative quantities is 1.09% in SS. Truncation changes welfare by about 0.13 percentage points at most, and the scenario ranking is unchanged.

Long-Run BGP Feasibility

The long-run analysis evaluates the BGP properties of the solved closed-loop systems. The main counterfactuals evaluate welfare and g_Y at the observed 2017 knowledge-capital distribution.

Baseline BGP. For the long-run BGP, the dominant eigenpair of the unconstrained closed-loop system solves

$$\boldsymbol{\Phi}\bar{\boldsymbol{z}} = g\bar{\boldsymbol{z}}, \qquad \boldsymbol{\Phi} = \boldsymbol{\Omega} - \delta \boldsymbol{I} + \mu^2 \widetilde{\boldsymbol{X}}. \tag{49}$$

Although $\boldsymbol{\Omega}$ is nonnegative under the row-receiver spillover convention, $\widetilde{\boldsymbol{X}}$ contains strategic R&D policy feedbacks and can have negative off-diagonal elements. Hence $\boldsymbol{\Phi}$ is not generally a Metzler matrix and need not preserve the positive orthant; a positive

Table 20: Transition-Path Sign Violations and Welfare Effect of Truncating Negative R&D Effort at Zero

Scenario	First $x < 0$ (years)	First $q < 0$ (years)	Max cost share from $x < 0$ (%)	Max q^2 share from $q < 0$ (%)	Welfare effect of truncation (pp)
CC	24.5	33.3	0.004	0.002	0.0001
CM	0.0	30.9	6.617	0.004	0.1321
CS	0.0	23.9	0.084	0.011	-0.0005
MM	0.0	0.0	0.954	0.125	0.0196
SS	0.0	0.0	0.764	1.086	-0.0115
33% subsidy	15.4	26.9	0.012	0.007	0.0001

Notes: The sign columns simulate deterministic closed-loop transition paths for 50 years from the observed 2017 knowledge-capital vector; the 33% subsidy row uses the welfare-maximizing uniform-subsidy equilibrium. Knowledge capital remains positive in every scenario. A value of 0.0 in a first-violation column means the sign violation is present at the observed state. Cost share from $x < 0$ is $\sum_{i:x_i<0} x_i^2 / \sum_i x_i^2$; the q^2 share from $q < 0$ is defined analogously. The final column reports the welfare change, in percentage points of the untruncated CC value, after negative R&D effort is set to zero in both the knowledge drift and resource cost, holding the feedback coefficients fixed. The calculation integrates household payoffs for 100 years and adds a fixed-active-set terminal tail whose share is below 0.043 percent in every row.

Table 21: Positive-State Feasibility of the Unconstrained Dominant Balanced Growth Path (BGP)

Scenario	Neg. $\bar{z}$	Neg. $\lvert\bar{z}\rvert$ share (%)
CC	508/757	47.6
CM	410/757	49.4
CS	356/757	49.5
MM	443/757	49.3
SS	297/757	50.9

Notes: The dominant eigenvector is oriented toward the observed 2017 knowledge-capital vector and normalized by $\sum_i |\bar{z}_i| = 1$. Neg. $\bar{z}$ counts negative entries; Neg. $|\bar{z}|$ share reports their absolute mass.

dominant eigenvector is therefore not guaranteed by Perron–Frobenius arguments. After orientation toward the observed knowledge-capital vector and normalization by $\sum_i |\bar{z}_i| = 1$, $\bar{z}$ is the candidate BGP distribution. A positive BGP requires $\bar{z}_i > 0$ for every firm. Table 21 reports the number and mass of violations in the five counterfactual scenarios.

Technology spillover floor. The baseline BGP feasibility results show that the unconstrained dominant eigenvectors are mixed-sign in all five 2017 scenarios. To gauge whether this long-run sign issue is driven by the sparse allocation of empirical spillover links, we compute one perturbation for the competitive benchmark. The perturbation preserves each recipient firm's total spillover exposure but spreads a fraction λ uniformly across other source firms:

$$\Omega_{ij}^{(\lambda)} = (1-\lambda)\Omega_{ij} + \lambda \frac{\sum_{k \neq i} \Omega_{ik}}{n-1}, \qquad j \neq i, \tag{50}$$

with the diagonal unchanged. For $\lambda = 0.15$, the oriented dominant eigenvector of the competitive benchmark is positive. The perturbation changes the instantaneous drift by only about 2.7 percent in relative ℓ_2 norm and leaves deterministic transitions from the observed 2017 state nearly unchanged. Over a 50-year horizon, the cross-firm correlation of knowledge-growth rates with the baseline path stays above 0.996, aggregate R&D remains within 1 percent of the baseline path, and the continuation value at the observed state, $V(\boldsymbol{z}_0) = \boldsymbol{z}_0^T \boldsymbol{X} \boldsymbol{z}_0$, differs by only 0.4 percent. This exercise is not a re-estimated model and is not used in the main counterfactuals; it only establishes that the observed-state comparison is not mechanically driven by the mixed-sign asymptotic BGP.